

Statutory Incidence of Ad Valorem Taxes: Revisiting Classical Theory and Policy Implications

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Abstract

This paper revisits the canonical result that statutory tax incidence is theoretically irrelevant for economic outcomes in competitive equilibrium. Shifting statutory incidence toward the demand side reduces the consumer price, raises the supplier price, and increases quantity, whereas the tax revenue effect is ambiguous. Changes in the tax base and effective tax rates explain these results. We derive new empirically tractable incidence formulas in response to statutory incidence shifts, which we quantify using simulations in the payroll tax context. We apply these formulas to reduced form estimates of price changes after statutory incidence shifts, allowing us to point-identify elasticities and parameters like tax evasion. Finally, we apply our framework to OECD payroll taxes, documenting that employer shares rise with total payroll taxes and offset nominal tax differences. These stylized facts further highlight the importance of accounting for statutory incidence in policy and empirical analyses.

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1 Introduction

A canonical result in public finance holds that the economic incidence of a tax is independent of its statutory (or legal) incidence—absent frictions and in competitive equilibrium. Public finance textbooks (e.g., [Dalton, 1941](#); [Salanié, 2003](#); [Gruber, 2019](#)) and handbook chapters (e.g., [Kotlikoff and Summers, 1987](#); [Fullerton and Metcalf, 2002](#)) discuss this extensively:

“[I]t is clear that the determination of the equilibrium quantity, the price paid by consumers, and the net of tax receipts of producers does not depend on which side of the market the tax is levied.” ([Kotlikoff and Summers, 1987](#))

This theoretical irrelevance has motivated extensive empirical research, which generally finds that statutory incidence affects economic outcomes (see [Benzarti, 2025](#)). For example, [Jiménez et al. \(2024\)](#) study a statutory shift of mortgage taxes in Spain—an ad valorem tax—from borrowers to banks without changing the tax rate, motivated by the “key principle [...] that (economic) tax incidence [...] is independent of on which agent taxes are levied [...] as price adjustments compensate such shift.” In tax design, however, statutory incidence plays a central role. Payroll taxes, for instance, are typically divided nominally between the employer and employee, consistent with a policy view that who formally pays the tax (statutory incidence) also bears it economically (economic incidence).

This paper reexamines this classical irrelevance result by introducing statutory incidence as an explicit policy parameter into the study of economic incidence. Unlike commonly asserted, we show that statutory incidence affects economic incidence for ad valorem taxes.¹ We apply and discuss these theoretical insights to payroll taxes among OECD countries. We further apply our new incidence formulas to the empirical setting in [Bibler et al. \(2021\)](#)—who study a statutory shift of taxes on Airbnb rentals from hosts to renters—to point-identify supply or demand elasticities and tax evasion.

We first clarify how statutory incidence matters for ad valorem taxes by defining the concepts of strong and weak irrelevance. Statutory incidence is *strongly* irrelevant for economic incidence if shifting the statutory incidence at a constant tax rate keeps prices, quantity, and tax revenue unchanged. In contrast, statutory incidence is *weakly* irrelevant if shifting the statutory incidence maintains the economic equilibrium and collected revenue whilst allowing for simultaneous tax rate adjustments. While per unit taxes satisfy strong and weak irrelevance, ad valorem taxes only satisfy weak irrelevance.

Consider an ad valorem tax τ_0 fully levied on the demand side such that $p_D^0 = (1 + \tau_0)p_S^0$. Suppose a reform shifts the statutory incidence onto the supply side at constant τ_0 . Prices adjust in equilibrium to equate supply and demand at $p_S^1 = (1 - \tau_0)p_D^1$. Unlike in the

¹Ad valorem taxes are proportionate to the price of the good, i.e., $(1 + \tau)p$. In comparison, a per unit tax—also called specific tax—is a constant amount per unit of the good irrespective of its price, i.e., $p + t$.

per unit tax case, however, the price adjustment does not neutralize the statutory incidence shift, changing real economic allocation, tax revenue, and incidence. Under general conditions—downward-sloping demand, upward-sloping supply—shifting statutory incidence at constant ad valorem tax rate toward the demand side increases the equilibrium quantity, raises supply prices, and lowers demand prices. The revenue effect is ambiguous and depends on the elasticities, but increases unambiguously for shifts toward the supply side when $|\varepsilon^D| < 1$.

Strong irrelevance fails for ad valorem taxes due to differences in the tax base. When the tax is levied on the demand side, the tax base is the supply price, p_S . This creates a wedge of $\tau_0 p_S$ between the demand and supply curve. When the tax is levied on the supply side, the tax base is the demand price, p_D , creating a wedge of $\tau_0 p_D$. Since $p_S \leq p_D$ for $\tau_0 \geq 0$, the distortion is smaller when τ_0 is levied on the demand side. Importantly, an ad valorem tax of τ_0 levied on the supply side is economically different from an ad valorem tax of τ_0 on the demand side. As [Fullerton and Metcalf \(2002\)](#) discuss, a 50% demand-side tax that raises the price from \$1 to \$1.50 corresponds to a supply-side tax of 33% on \$1.50. This correspondence demonstrates the weak statutory irrelevance for ad valorem taxes, which we generalize to $\tau_1 = \frac{\tau_0}{1 - d\alpha\tau_0}$ for arbitrary τ_0 and $d\alpha$.

We introduce new empirically tractable formulas for the economic incidence of shifts in the statutory incidence. These formulas closely mirror canonical economic incidence formulas for tax rate changes and depend on relative demand and supply elasticities, the tax rate, and—importantly—the statutory incidence. These formulas highlight that changes in statutory incidence for ad valorem taxes implicitly act as effective tax rate changes. We extend these incidence formulas to capture frictions like tax salience (e.g. [Chetty et al., 2009](#)) or tax evasion (e.g. [Kopczuk et al., 2016](#)). We also derive general economic incidence formulas for tax rate changes accounting for statutory incidence, reinforcing that incorporating statutory incidence is important for ad valorem tax rate changes.

We then simulate the economic incidence of statutory incidence shifts and tax rate changes in the payroll tax context. These results emphasize how statutory incidence affects expected price changes. Statutory incidence shifts at constant tax rates can lead to meaningful price changes for medium and large tax rates common for payroll taxes. For instance, solely shifting the statutory incidence from the supply to the demand side at $\tau = 0.25$, $\varepsilon^S = 0.1$, and $\varepsilon^D = -1$, leads to $\frac{dp_S}{p_S} = -7.6\%$ and $\frac{dp_D}{p_D} = 0.7\%$. This has implications for (i) tax policy and (ii) empirical analyses of tax reforms, which must distinguish predicted price changes in the frictionless, competitive benchmark from other factors like salience or evasion.

We demonstrate the policy relevance of these theoretical insights for payroll taxes among OECD countries. We establish three stylized facts. First, accounting for differences in

statutory incidence is important for comparing payroll tax rates across countries. Second, total payroll taxes and employer shares correlate positively, implying more muted cross-country differences in effective than nominal payroll tax rates. Third, this relationship holds within types of payroll taxes or controlling for country and year fixed effects. This pattern is consistent with a “traditional” policymaker’s intention to reduce the burden on workers by shifting the statutory incidence toward employers. Our theory suggests this policy achieves the intended effect of lowering workers’ tax burden indirectly through lowering effective tax rates on both sides.

Finally, we demonstrate the empirical applicability of our new formulas for $\frac{dp}{d\alpha}$, building on the results in [Bibler et al. \(2021\)](#). We show how our formulas can point-identify elasticities and additional parameters like tax evasion. [Bibler et al. \(2021\)](#) study a shift in the statutory incidence and remittance of taxes on Airbnb stays from hosts to renters (remitted through Airbnb). They provide an upper bound of pre-reform tax compliance by hosts of 24% and a lower bound for the supply elasticity of 1.5. Our new incidence formulas enable us to refine their results, resulting in a pre-reform tax compliance of 0.8%, consistent with the complementary methodology developed in [Bibler et al. \(forthcoming\)](#).

Contribution to the literature. We contribute to the theory of tax incidence, empirical tax incidence studies, and the role of statutory incidence as a tax policy instrument.

Our first theoretical contribution is to incorporate statutory incidence as an explicit, continuous policy parameter, α , in the study of tax incidence. This allows us to (i) derive new, empirically tractable incidence formulas for statutory incidence changes, $\frac{dp}{d\alpha}$ and $\frac{dQ}{d\alpha}$, (ii) show that $\frac{dp}{d\alpha} \neq 0$ for ad valorem taxes even in the competitive and frictionless benchmark, and (iii) simulate the magnitude of this novel statutory incidence channel. Our second theoretical contribution is the introduction of strong and weak statutory irrelevance.

This contributes to substantial theoretical work on ad valorem taxes. For instance, [Musgrave \(1959\)](#) provides derivations of $\frac{dp}{d\tau}$ for ad valorem taxes on the supply ($\alpha = 0$) and the demand ($\alpha = 1$) side and presents their equivalence at $\tau_s = \frac{\tau_d}{1+\tau_d}$. [Musgrave](#) further discusses that revenue-equivalent ad valorem and per unit taxes lead to the same equilibrium. Other work also highlights this correspondence $\tau = \frac{t}{p}$ for equivalent ad valorem and per unit taxes (e.g., [Fullerton and Metcalf, 2002](#); [Hindriks and Myles, 2013](#); [Stiglitz and Rosengard, 2015](#)). We generalize these results for $\alpha \in [0, 1]$. Consistent with the prior literature, [Pauwels and Schroyen \(2024\)](#) demonstrate that increasing ad valorem taxes on the supply side leads to larger price changes than tax rate increases on the demand side. [Hargaden \(2026\)](#) demonstrates that U.S. tax credits on foreign payroll taxes provide an application of our core results on how statutory incidence affects economic incidence.

Our next theoretical contribution is to incorporate frictions, such as tax evasion, into our

incidence formulas. This relates to empirical work on tax incidence, which generally finds that statutory incidence matters in the presence of frictions such as tax salience (Chetty et al., 2009), differential evasion and enforcement (Kopczuk et al., 2016; Slemrod, 2019; Hargaden and Roantree, 2019; Bibler et al., 2021; Fox et al., 2022; Garriga and Tortarolo, 2024), price rigidities and norms (Saez et al., 2012), imperfect competition (Hansen et al., 2026), and bargaining (Claussen et al., 2026; Jiménez et al., 2024). Synthesizing this literature, Benzarti (2025) discusses various “tax anomalies”, outlining key shortcomings of the canonical incidence model.²

Nevertheless, recent empirical work on statutory incidence shifts of ad valorem taxes is motivated by the “conventional wisdom” that statutory incidence changes should not affect prices or quantities in the competitive, frictionless benchmark (e.g. Bibler et al., 2021; Jiménez et al., 2024; Claussen et al., 2026). We argue this “conventional wisdom” possibly emerged for two reasons.³ First, algebraic incidence results of per unit taxes are commonly transferred to ad valorem taxes without further proof (e.g., Dalton, 1941; Kotlikoff and Summers, 1987). Second, algebraic results for ad valorem taxes only consider the introduction of an infinitesimal tax at initial $\tau = 0$ (e.g., Fullerton and Metcalf, 2002; Salanié, 2003). As we show, the statutory incidence is indeed irrelevant when introducing an infinitesimal ad valorem tax at $\tau = 0$; the irrelevance fails, however, for any given non-zero ad valorem tax, which is the relevant case for most settings. Our new incidence formulas contribute to this literature by providing a clear and empirically applicable framework that correctly incorporates statutory incidence and allows for frictions.

Beyond providing empirically estimable incidence formulas, we contribute to the empirical tax incidence literature by showing how our formulas for $\frac{dp}{d\alpha}$ can identify demand or supply elasticities, as well as other parameters. We apply our formulas to the reduced-form estimates in Bibler et al. (2021) and refine their results showing that pre-reform tax compliance of Airbnb hosts in the US was near zero (0.8%) when the tax was levied on hosts. These results align with the 0–1.6% pre-reform compliance found in Bibler et al. (forthcoming), who develop an alternative methodology using structural log-linear demand and supply functions—building on Zoutman et al. (2018)—to point-identify tax evasion and elasticities following various tax enforcement changes, including statutory incidence shifts.

²Saez and Zucman (2024) propose a new framework assigning the economic incidence of a tax by its base. Bozio et al. (2026) provide compelling evidence that the tax–benefit linkage of payroll taxes matters for economic incidence, and that in the absence of such a linkage, economic incidence coincides with statutory incidence.

³This “conventional wisdom” is typically presented for a competitive economy. The classical irrelevance rests on the assumption that all information to determine tax liability is freely and universally observed (Slemrod, 2008). Musgrave (1959), among others, shows how an equivalence between per unit and ad valorem taxes breaks down under imperfect competition. See Weyl and Fabinger (2013) for an overview of tax incidence under imperfect competition.

Finally, we demonstrate the importance of the statutory incidence channel for policymaking and analyses in the payroll tax context. We provide new stylized facts that are consistent with policymakers setting statutory incidence with economic incidence in mind. These findings also imply that effective tax rates vary less across countries than nominal rates. Our results caution that isolated shifts in statutory incidence are implicit tax rate changes with the risk of unintentionally raising effective tax rates—as seen in recent payroll tax changes.

We proceed as follows. Section 2 introduces notation and defines strong and weak statutory irrelevance. Section 3 recalls the strong irrelevance of per unit taxes. Section 4 generalizes and discusses the failure of strong irrelevance of ad valorem taxes. Section 5 demonstrates how weak irrelevance of ad valorem taxes can be restored. Section 6 provides a policy application to payroll taxes in OECD countries and section 7 an empirical application to a statutory incidence shift of Airbnb taxes. Section 8 concludes.

2 Notation and definitions

Let p denote the tax-exclusive price of some good and p_S and p_D the tax-inclusive prices on the supply and demand side. Denote $S(p_S)$ and $D(p_D)$ as aggregate supply and demand. Express per unit taxes by t and ad valorem taxes by τ . The statutory incidence of a given tax may be split across the demand and supply sides; denote $\alpha \in [0, 1]$ as the statutory share of the tax falling on the demand side. Thus:

$$\begin{aligned} \text{per unit tax:} \quad & p_D = p + \alpha t \quad \text{and} \quad p_S = p - (1 - \alpha)t \\ \text{ad valorem tax:} \quad & p_D = (1 + \alpha\tau)p \quad \text{and} \quad p_S = [1 - (1 - \alpha)\tau]p, \end{aligned}$$

where the tax-exclusive price p adjusts in equilibrium to equate supply and demand.

To isolate the role of statutory incidence, we abstract from tax remittance (Slemrod, 2008) and from tax–benefit linkages (Bozio et al., 2026), which can matter in practice.

Strong and weak statutory irrelevance

We consider the statutory incidence of some tax to be *strongly irrelevant* for economic incidence whenever real economic outcomes are independent of the statutory incidence holding the tax rate constant (Definition 1). In contrast, we define the statutory incidence to be *weakly irrelevant* whenever the real economic outcomes are unaffected by the statutory incidence whilst allowing for a corresponding adjustment in the tax rate (Definition 2). Strong irrelevance implies weak irrelevance, but not vice versa.

Definition 1 (Strong irrelevance of statutory incidence). *The statutory incidence of some tax regime $\mathbb{T} = \{T, \alpha\}$ with $T \in \{t, \tau\}$ is strongly irrelevant if for all $\alpha_0, \alpha_1 \in [0, 1]$ and any $T_0 \in \mathbb{R}$*

we have

- Constant prices: $p_S(T_0, \alpha_0) = p_S(T_0, \alpha_1)$ and $p_D(T_0, \alpha_0) = p_D(T_0, \alpha_1)$
- Constant quantity: $Q(T_0, \alpha_0) = Q(T_0, \alpha_1)$
- Constant revenue: $R(T_0, \alpha_0) = R(T_0, \alpha_1)$

Definition 2 (Weak irrelevance of statutory incidence). *The statutory incidence of some tax regime $\mathbb{T} = \{T, \alpha\}$ with $T \in \{t, \tau\}$ is weakly irrelevant if for all $\alpha_0, \alpha_1 \in [0, 1]$ and any $T_0 \in \mathbb{R}$ there exists $T_1 \in \mathbb{R}$ such that*

- Constant prices: $p_S(T_0, \alpha_0) = p_S(T_1, \alpha_1)$ and $p_D(T_0, \alpha_0) = p_D(T_1, \alpha_1)$
- Constant quantity: $Q(T_0, \alpha_0) = Q(T_1, \alpha_1)$
- Constant revenue: $R(T_0, \alpha_0) = R(T_1, \alpha_1)$

Under strong irrelevance, the economic incidence of a tax rate change is independent of its statutory incidence (**Corollary 1**).

We frequently express formulas in terms of tax-inclusive price elasticities. Formally, these are $\varepsilon^S := \frac{\partial S(p_S)}{\partial p_S} \frac{p_S}{S(p_S)}$ and $\varepsilon^D := \frac{\partial D(p_D)}{\partial p_D} \frac{p_D}{D(p_D)}$. Throughout, we assume $\varepsilon^S \geq 0$ and $\varepsilon^D \leq 0$.

3 Strong irrelevance of per unit taxes

To contrast ad valorem taxes, this section rederives the well-known insight that the statutory incidence of per unit taxes is irrelevant for economic incidence in a frictionless, competitive environment.

3.1 Strong irrelevance

Consider a per unit tax t for $\alpha \in (0, 1)$.⁴ Proposition 1 establishes the strong statutory irrelevance of per unit taxes, implying economic incidence is independent of statutory incidence (Corollary 1).

Proposition 1. *Per unit taxes satisfy strong irrelevance of statutory incidence.*

Proof of proposition 1. Totally differentiating both sides of $S(p - (1 - \alpha)t) = D(p + \alpha t)$ with $d\alpha \neq 0$ and $dt = 0$, we obtain $\frac{\partial S}{\partial p_S} [dp + t d\alpha] = \frac{\partial D}{\partial p_D} [dp + t d\alpha]$, which is equivalent to $\left[\frac{\partial S}{\partial p_S} - \frac{\partial D}{\partial p_D} \right] [dp + t d\alpha] = 0$. Hence, unless $\frac{\partial S}{\partial p_S} = \frac{\partial D}{\partial p_D} = 0$, we must have $dp = -t d\alpha$.⁵ From $dp_D = dp + t d\alpha$ and $dp_S = dp + t d\alpha$, it follows $dp_D = dp_S = 0$. Finally, $dS = S'(p_S) dp_S$ and $dD = D'(p_D) dp_D$ imply $dQ^* = 0$ and $dR^* = dQ^* t + dt Q^* = 0$, satisfying Definition 1. $\square$

⁴All insights on statutory irrelevance hold for $\alpha = 0$ or $\alpha = 1$, respectively.

⁵Since $\frac{\partial S}{\partial p_S} \geq 0$ and $\frac{\partial D}{\partial p_D} \leq 0$ by assumption, the only possibility for $\left(\frac{\partial S}{\partial p_S} - \frac{\partial D}{\partial p_D} \right) = 0$ is that $\frac{\partial S}{\partial p_S} = 0$ and $\frac{\partial D}{\partial p_D} = 0$.

3.2 Economic incidence formulas

The economic incidence of a tax is often expressed as the change in (tax-inclusive) prices in response to a change in the tax. Specifically, a per unit tax reform with $dt \neq 0$ and $d\alpha = 0$ for arbitrary $\alpha \in [0, 1]$ leads to the following price changes:

$$\frac{dp_S}{dt} = \frac{\varepsilon^D - \frac{\partial D}{\partial p_D} \frac{t}{D}}{\varepsilon^S - \varepsilon^D + \frac{\partial D}{\partial p_D} \frac{t}{D}} \quad (1)$$

$$\frac{dp_D}{dt} = \frac{\varepsilon^S}{\varepsilon^S - \varepsilon^D + \frac{\partial D}{\partial p_D} \frac{t}{D}}. \quad (2)$$

Online Appendix B.1 provides a detailed derivation of this result, which is obtained from totally differentiating $S(p - (1 - \alpha)t) = D(p + \alpha t)$ with $d\alpha = 0$ and $dt \neq 0$ to obtain an expression for $\frac{dp}{dt}$, and totally differentiating the expressions for p_S and p_D . The introduction of a small per unit tax, i.e., $t = 0$ and $dt \neq 0$, depicts the classical textbook formula:

$$\frac{dp_S}{dt} = \frac{\varepsilon^D}{\varepsilon^S - \varepsilon^D} \quad \text{and} \quad \frac{dp_D}{dt} = \frac{\varepsilon^S}{\varepsilon^S - \varepsilon^D}.$$

The strong irrelevance result implies prices are unaffected by shifts in the statutory incidence, i.e., $\frac{dp_S}{d\alpha} = 0$ and $\frac{dp_D}{d\alpha} = 0$. Accordingly, equations (1) and (2) remain valid for price changes due to a small tax policy reform in both the statutory incidence and the tax rate.

4 Failure of strong irrelevance of ad valorem taxes

We now turn to the case of ad valorem taxes. Section 4.1 formally proves the failure of strong irrelevance. Section 4.2 provides intuition for this result. We then derive economic incidence formulas in section 4.3 for changes in the statutory incidence and tax rate. Section 4.4 demonstrates the magnitude of the statutory incidence channel in numerical examples.

4.1 Formal proof

Proposition 2 shows that ad valorem taxes fail strong statutory irrelevance (Definition 1). In turn, the statutory incidence of a given ad valorem tax rate *is* relevant for real economic outcomes and the economic incidence of the tax.

Proposition 2. *Ad valorem taxes do not satisfy strong irrelevance of statutory incidence.*

Proof of proposition 2. Definition 1 requires $\frac{dp_S}{d\alpha} = \frac{dp_D}{d\alpha} = 0$. We will prove that $\frac{dp_S}{d\alpha} = 0$ and $\frac{dp_D}{d\alpha} = 0$ cannot hold simultaneously when $\tau \neq 0$. First, consider $p_S = [1 - (1 - \alpha)\tau]p$. Totally differentiating both sides, and imposing $d\tau = 0$, $\frac{dp_S}{d\alpha} = [1 - (1 - \alpha)\tau]\frac{dp}{d\alpha} + \tau p$.

Postulating $\frac{dp_S}{d\alpha} = 0$, re-arrange to obtain $\frac{dp}{p} = -\tau d\alpha \frac{1}{1-(1-\alpha)\tau}$. Second, consider $p_D = (1 + \alpha\tau)p$. Totally differentiating and imposing $d\tau = 0$, $\frac{dp_D}{d\alpha} = (1 + \alpha\tau)dp + \tau p d\alpha$. Again, postulating $\frac{dp_D}{d\alpha} = 0$, re-arrange to obtain $\frac{dp}{p} = -\tau d\alpha \frac{1}{1+\alpha\tau}$. For these equations to hold simultaneously, we need $\tau = 0$. Thus, for any $\tau \neq 0$, Definition 1 is not satisfied. $\square$

4.2 Intuition

The distortion of an ad valorem tax depends on the tax rate *multiplied* by its tax base—the tax-exclusive price p . Strong irrelevance fails for ad valorem taxes because a shift in the statutory incidence changes the *base* without adjusting the tax rate. While the tax base, p , adjusts to statutory incidence changes to maintain an equilibrium, it does not fully offset the statutory incidence shift (unlike in the per unit tax case).

Consider a tax levied fully on the supply side, which is shifted fully on the demand side. What happens in equilibrium? (see Figure A.1 for graphical intuition.)

1. Prior to the tax reform, $p_S^0 = (1 - \tau_0)p_D^0$. The tax creates a wedge equal to $\tau_0 p_D$.
2. The reform shifts the tax entirely on the demand side. Consider the (hypothetical) market immediately after the reform before the supplier adjusts the price. Then, $\tilde{p}_D = (1 + \tau_0)p_S$. Since $p_S^0 < p_D^0$, the wedge is smaller and consumers face a lower price $\tilde{p}_D < p_D^0$ with excess demand $D(\tilde{p}_D) > S(p_S^0)$.
3. The supplier raises the price until demand equals supply. Now, $p_D^1 = (1 + \tau_0)p_S^1$ with $p_S^1 > p_S^0$. Since $p_S^1 \leq p_D^0$ for $\tau \geq 0$, the wedge is less distortive at $\tau_0 p_S^1 < \tau_0 p_D^0$.

4.3 Economic incidence formulas

Canonical economic incidence analyses ask how tax-inclusive prices adjust when the tax rate changes. Here, we extend this analysis to changes in (i) the statutory incidence at constant tax rate and (ii) the tax rate at constant statutory incidence—with and without frictions.⁶ Online Appendix B.2 provides detailed derivations of these results.

Economic incidence of a change in statutory incidence

Suppose a tax reform shifts the statutory incidence by $d\alpha$ while keeping the tax rate unchanged. What are the effective changes in tax-inclusive prices p_S and p_D ? Result 1 first summarizes how tax-exclusive p changes in response to the reform $d\alpha$.

Result 1 (Tax-exclusive price change due to statutory incidence shifts). *Consider a tax-reform*

⁶The analysis considers small changes; simultaneous variations in the tax rate and in statutory incidence are additively separable.

that shifts the statutory incidence by $d\alpha$ without altering the ad valorem tax rate, i.e., $d\tau = 0$. Then,

$$\frac{dp}{p} = -\tau d\alpha \frac{\frac{\varepsilon^S}{1-(1-\alpha)\tau} - \frac{\varepsilon^D}{1+\alpha\tau}}{\varepsilon^S - \varepsilon^D}. \quad (3)$$

Result 1 follows from totally differentiating $S([1 - (1 - \alpha)\tau]p) = D((1 + \alpha\tau)p)$ with $d\tau = 0$, using $S = D$, the definitions of ε^S and ε^D , and the relationships between p_S , p_D , and p .

To shed light on the economic incidence of the statutory incidence shift, result 2 describes tax-inclusive price changes. The expressions follow from totally differentiating $p_S = [1 - (1 - \alpha)\tau]p$ and $p_D = (1 + \alpha\tau)p$, the result for $\frac{dp}{p}$, and setting $d\tau = 0$.

Result 2 (Tax-inclusive price changes due to statutory incidence shifts). *Consider a tax-reform that shifts the statutory incidence by $d\alpha$ without altering the ad valorem tax rate, i.e., $d\tau = 0$. Then,*

$$\frac{dp_S}{p_S} = d\alpha \frac{\tau^2}{(1 + \alpha\tau)[1 - (1 - \alpha)\tau]} \frac{-\varepsilon^D}{\varepsilon^S - \varepsilon^D} \quad (4)$$

$$\frac{dp_D}{p_D} = d\alpha \frac{\tau^2}{(1 + \alpha\tau)[1 - (1 - \alpha)\tau]} \frac{-\varepsilon^S}{\varepsilon^S - \varepsilon^D}. \quad (5)$$

Result 2 has important economic implications. First, the incidence is proportional to the relative elasticities, and falls more on the less elastic side. Second—which is worth stressing—the price change is non-zero since ad valorem taxes do not satisfy strong irrelevance. Third, for $\alpha \neq 1$, a reform that only shifts the statutory incidence towards the demand side ($d\alpha > 0$ and $d\tau = 0$) increases the supplier price ($\frac{dp_S}{p_S} > 0$) and decreases the consumer price ($\frac{dp_D}{p_D} < 0$). As a corollary—for a given tax rate τ_0 —the equilibrium quantity is maximized when the statutory incidence falls fully on the demand side (**Corollary 2**).

Since tax revenue is $R = \tau pQ$, and p and Q move in opposite directions for statutory incidence changes, $d\alpha \neq 0$, the revenue-maximizing α at constant τ is ambiguous. However, for $\varepsilon^D \geq -1$, revenue is maximized when $\alpha = 0$, i.e., the tax falls fully on the supply side (**Corollary 3**).⁷

Economic incidence of a change in statutory incidence with frictions

The frictionless framework can be easily extended to include frictions such as reduced tax salience (motivated by Chetty et al., 2009) or tax evasion (motivated by Kopczuk et al., 2016). We focus on supply-side tax evasion here (to match our empirical application) and provide demand-side salience in Online Appendix B.4.2. We add a tax evasion parameter $\gamma \in [0, 1]$

⁷Totally differentiating $R = \tau pQ = \tau pD$ gives $\frac{dR}{R} = \frac{dp}{p} + \frac{dD}{D}$. This can be rewritten as $dR = (1 + \varepsilon^D) \frac{dp}{p} + \frac{\tau}{1+\alpha\tau} \varepsilon^D d\alpha$.

in the expression of tax-inclusive prices

$$\begin{aligned} p_D &= (1 + \alpha\tau)p \\ p_S &= [1 - (1 - \alpha)\gamma\tau]p. \end{aligned}$$

Result 3 characterizes expected price changes in the presence of this supply side friction. Importantly, tax revenue would be lower with $R = [\alpha\tau + \gamma(1 - \alpha)\tau]pQ < \tau pQ$.

Result 3 (Tax-inclusive price changes due to statutory incidence shifts in the presence of supply-side frictions). *Consider a tax-reform that shifts the statutory incidence by $d\alpha$ without altering the ad valorem tax rate, i.e., $d\tau = 0$. Then,*

$$\frac{dp_S}{p_S} = \tau d\alpha \frac{1 - \gamma\tau - \gamma}{(1 - (1 - \alpha)\gamma\tau)(1 + \alpha\tau)} \frac{\varepsilon^D}{\varepsilon^S - \varepsilon^D} + \tau d\gamma \frac{1 - \alpha}{1 - (1 - \alpha)\gamma\tau} \frac{\varepsilon^D}{\varepsilon^S - \varepsilon^D} \quad (6)$$

$$\frac{dp_D}{p_D} = \tau d\alpha \frac{1 - \gamma\tau - \gamma}{(1 - (1 - \alpha)\gamma\tau)(1 + \alpha\tau)} \frac{\varepsilon^S}{\varepsilon^S - \varepsilon^D} + \tau d\gamma \frac{(1 - \alpha)}{1 - (1 - \alpha)\gamma\tau} \frac{\varepsilon^S}{\varepsilon^S - \varepsilon^D} \quad (7)$$

For simplicity, the following example assumes $d\gamma = 0$. Note that $\frac{1 - \gamma\tau - \gamma}{(1 - (1 - \alpha)\gamma\tau)(1 + \alpha\tau)}$ is negative for sufficiently high tax rates τ and low evasion levels (high γ). In this case, a statutory incidence shift toward the demand side ($d\alpha > 0$) raises the supply price and lowers the demand price, consistent with standard results for statutory incidence changes. However, when evasion is sufficiently high (low γ), the pattern reverses: the supply price falls and the demand price rises because the direct mechanical incidence effect is dominated by an effective increase in the tax burden as the statutory shift improves tax compliance.

Economic incidence of a change in the tax rate

Consider now a tax reform that changes the tax rate by $d\tau$ while keeping the statutory incidence unchanged. How will the tax-inclusive prices p_S and p_D change?

Result 4 (Tax-inclusive price changes due to tax rate changes). *Consider a tax-reform that changes the tax rate by $d\tau \neq 0$ while keeping the statutory incidence unchanged, i.e., $d\alpha = 0$. Then, tax-inclusive prices change by*

$$\frac{dp_S}{p_S} = d\tau \frac{1}{[1 - (1 - \alpha)\tau](1 + \alpha\tau)} \frac{\varepsilon^D}{(\varepsilon^S - \varepsilon^D)} \quad (8)$$

$$\frac{dp_D}{p_D} = d\tau \frac{1}{[1 - (1 - \alpha)\tau](1 + \alpha\tau)} \frac{\varepsilon^S}{(\varepsilon^S - \varepsilon^D)}. \quad (9)$$

Equations (8) and (9) are similar to the well-established incidence formulas of per unit taxes expressed in equations (1) and (2). In both cases, the economic incidence is proportional

to the relative elasticities. Unlike the per unit tax case, however, the statutory incidence (α) matters for the economic incidence through the term $\frac{1}{[1-(1-\alpha)\tau](1+\alpha\tau)}$; this term only drops out when $\tau = 0$ reflecting the *introduction* of a small ad valorem tax. A further distinction—given the proportionate nature of an ad valorem tax—is that the economic incidence is expressed as a relative change. Finally, the ratio of relative price changes, $\left[\frac{dp_S}{p_S}\right] / \left[\frac{dp_D}{p_D}\right] = \frac{\varepsilon^D}{\varepsilon^S}$, is constant and independent of the statutory incidence and the tax rate, but empirical researchers are typically more interested in examining the separate effects on supply and demand prices, e.g., the effect on total labor costs or take-home pay.

4.4 Numerical examples of economic incidence

The previous section established that (i) shifts in the statutory incidence at constant tax rate will lead to changes in prices, and (ii) the statutory incidence matters for how prices change in response to changes in the tax rate. The following simulates the magnitude of these effects based on the incidence formulas derived in section 4.3.

We choose parameters to match the payroll tax context; p_S refers to the after-tax wage received by employees, and p_D refers to the per unit labor costs paid by employers. The only unknowns in the economic incidence formulas are the supply and demand elasticities, ε^S and ε^D , which we set based on the empirical literature: $\varepsilon^S \in [0.1, 0.3]$ (Chetty et al., 2011) and $\varepsilon^D \in [-1, -0.01]$ (Lichter et al., 2015).

Simulating economic incidence for a change in statutory incidence

Figure 1 panel I shows price changes, $\frac{dp_S}{p_S}$ and $\frac{dp_D}{p_D}$, in response to a tax reform that shifts the statutory incidence from the supply side ($\alpha = 0$) to the demand side ($d\alpha = 1$) at constant tax rate (see result 2). To interpret these price changes, recall that a shift of the statutory incidence from the supply to the demand side, at fixed nominal tax rate, acts like a decrease in the tax rate under the initial statutory incidence. Panel 1a depicts a case when supply is less elastic than demand. Consequently, the supply side bears a larger share of the economic incidence of the ad valorem tax. When the statutory incidence is shifted onto the demand side, the suppliers benefit proportionally more from the reduction in the effective tax rate. Panel 1b depicts the opposite case when demand is less elastic than supply.

The proportional price changes are small for low tax rates. For medium-to-large tax rates, however, the price changes are non-negligible. This has important implications for empirical research: when analyzing a reform with a meaningful shift in the statutory incidence and a sizable tax rate, prices *are* expected to change absent any frictions.

Simulating economic incidence for a change in tax rate

Figure 1 panel II shows expected price changes, $\frac{dp_S}{p_S}$ and $\frac{dp_D}{p_D}$, in response to a change in the tax rate. These price changes will depend on the statutory incidence, which we keep constant in the simulations (i.e., $d\alpha = 0$; see result 4). For illustrative purposes, we set $\tau_0 = 0.4$ and consider $d\tau = 0.05$.

An important insight of the above theory is that an ad valorem tax rate of τ_0 creates larger distortions when the statutory incidence falls on suppliers than when it falls on buyers. Thus, the price changes are smaller the more the statutory incidence falls on the demand side (α). As expected, the relative elasticities determine the degree to which suppliers or buyers bear the economic incidence.

These simulations reveal that the economic incidence of a nominal change in the tax rate differs substantially between a statutory incidence of $\alpha = 0$ and of $\alpha = 1$. When considering a tax rate change of $d\tau$, it is thus important to account for the statutory incidence of the tax.

5 Restoring weak irrelevance of ad valorem taxes

Section 4 established that ad valorem taxes fail the strong irrelevance of statutory incidence (definition 1). One interpretation of this result is that an ad valorem tax of τ_0 levied fully on the supply side is not the same “tax” as an ad valorem tax of τ_0 levied fully on the demand side. From a policy perspective, arguably, what matters is (i) how the tax distorts allocations and (ii) how much revenue it generates. This is captured in the notion of weak statutory irrelevance (definition 2): for an ad valorem tax at rate τ_0 on the supply side, is there a rate τ_1 that achieves the exact same outcomes when levied on the demand side?

Proposition 3 demonstrates that ad valorem taxes satisfy weak statutory irrelevance. The intuition is based on the observation that changes in statutory incidence and tax rates can offset each other. For each combination of statutory incidence and tax rate, there exist other combinations of statutory incidence *and* tax rate that result in the same equilibrium prices, quantity, and tax revenue. These different (τ, α) combinations are therefore equivalent to the same per unit tax that leads to the corresponding equilibrium.

Proposition 3. *Ad valorem taxes satisfy weak irrelevance of statutory incidence.*

Proof of proposition 3. Consider tax schedules $\mathbb{T}_0 = (\tau_0, \alpha_0)$ and $\mathbb{T}_1 = (\tau_1, \alpha_1)$ with

$$\begin{aligned} (\tau_0, \alpha_0) : \quad & p_D^0 = (1 + \alpha_0 \tau_0) p^0 \text{ and } p_S^0 = [1 - (1 - \alpha_0) \tau_0] p^0 \\ (\tau_1, \alpha_1) : \quad & p_D^1 = (1 + \alpha_1 \tau_1) p^1 \text{ and } p_S^1 = [1 - (1 - \alpha_1) \tau_1] p^1. \end{aligned}$$

Suppose prices are equalized under both regimes, $p_D^0 = p_S^1$ and $p_D^1 = p_S^0$. If this holds, then

$\mathbb{T}_0$ and $\mathbb{T}_1$ achieve equal equilibrium quantities. This condition is true for

$$\tau_1 = \frac{\tau_0}{1 + (\alpha_0 - \alpha_1)\tau_0} = \frac{\tau_0}{1 - d\alpha \tau_0}. \quad (10)$$

One can then show that (10) implies $R_0 = \tau_0 p^0 = \tau_1 p^1 = R_1$. $\square$

For example, suppose τ_0 is fully levied on the supply side (i.e., $\tau_S = \tau_0$, $\alpha_0 = 0$). Shifting the tax fully onto the demand side ($d\alpha = 1$) results in the same quantities, prices, and tax revenue when $\tau_D = \frac{\tau_S}{1 - \tau_S}$, or equivalently $\tau_S = \frac{\tau_D}{1 + \tau_D}$.

6 Application to payroll taxes in OECD countries

This section characterizes implications of our theoretical insights for payroll taxes among OECD countries. Payroll taxes—or social insurance contributions—demonstrate the importance of statutory incidence in practice. Payroll taxes are taxes on the earnings of employees to fund various social insurance schemes, such as public pensions, health, or unemployment insurance. The statutory share of these taxes falling on the employer varies from 8% in Lithuania to 95% in Estonia among the OECD countries in our sample in 2024.

The following analyses combine data on payroll taxes we collected from [OECD \(2025\)](#), government websites, and consulting firms.⁸ The design of payroll taxes differs widely across countries both in terms of their level and coverage as well as the statutory split between employers and employees. We focus on OECD countries for which social contributions funded through payroll taxes cover at least pension, health, and unemployment insurance—the most common components of social insurances. We exclude countries where these insurances are funded through general income taxes. This includes 23 out of the 38 OECD countries. While the following findings are stylized and may reflect institutional factors—including different tax–benefit linkages—they support our theoretical prediction that statutory incidence should be carefully incorporated into economic incidence analyses.

Fact I: Differences in statutory incidence affect payroll tax comparisons across countries, requiring a standardization of statutory incidences for more accurate comparisons

A key theoretical insight is that effective tax rates depend on the statutory incidence. Heterogeneity in statutory incidence across countries’ payroll taxes thus renders cross-country comparisons of nominal tax rates inadequate. Instead, one should standardize tax rates by converting them to a common statutory incidence following equation (10). The OECD, for instance, reports the “tax wedge,” corresponding to our employer-side standardization.⁹

⁸We use average employer and employee payroll tax rates for full-time employees at 100% of the average wage.

⁹The [OECD](#) defines the tax wedge to be “the ratio between the amount of taxes paid by an average single

Figure 2a shows two such revenue-equivalent conversions: one where the tax falls fully on the employer ($\alpha = 1$, black X) and one where the tax falls fully on the employee ($\alpha = 0$, red squares). The revenue-equivalent tax rate is above the nominal rate when falling on the employer and is below the nominal rate when falling on the employee. This adjustment meaningfully shifts country rankings. For instance, while France exhibits the highest nominal rate, its standardized rate is lower than that of Austria. Slovenia and Costa Rica have very similar statutory rates, but around 10pp different standardized rates.

Fact II: Total payroll tax rates and employer shares are positively correlated, implying more muted cross-country differences in effective payroll tax rates

Figure 2b shows the relationship between payroll tax rates and statutory incidences in 2024. Tax rates range from around $\tau = 0.15$ to almost $\tau = 0.5$. While the employer bears at least half of the statutory incidence in almost all countries, substantial variation remains. Overall, there is a positive association between the tax rate and the employer share.¹⁰ This implies that differences in effective rates across countries are generally smaller than differences in nominal rates.

While the previous analysis provides a cross-country comparison for 2024, changes in the tax rate and in employers' shares from 2000 to 2024 are also positively correlated. This also holds in a regression with country and year fixed effects, where a 10 percentage-point increase in the tax rate is associated with a 2.1 percentage-point higher employer share ($p < 0.001$).

Fact III: The positive relationship between payroll tax rates and the employer share persists within types of payroll taxes

For better comparability and to identify specific patterns, Figure 2c decomposes payroll taxes into their specific components. We include the 16 countries that have separate rates on each of health insurance, pension, and unemployment insurance. The positive relationship between the tax rate and the employer share persists within each type. This suggests that the positive association for payroll taxes is not just the result of aggregating various payroll taxes of different sizes and employer shares.

Interpreting the facts: A policymaker's view and its implications on effective tax rates

Under the classical view, statutory incidence is irrelevant for economic incidence, so one might not expect any systematic correlation. By contrast, a "traditional" view equates statutory and economic incidence, leading policymakers to increase employer shares to reduce workers' tax burden. Empirically, average employer shares are lower for health

worker [...] and the corresponding total labour cost for the employer."

¹⁰All of the results in Figure 2 and those mentioned in this section are robust to excluding Lithuania (an outlier with a low tax rate and statutory incidence) or in the subset of countries shown in Figure 2c.

insurance (0.55) and pensions (0.57) than for unemployment insurance (0.67) and accident insurance (almost always 100%), consistent with the “traditional” view that employers bear greater responsibility for unemployment and workplace risks.

Recent payroll tax reforms illustrate this “traditional” view which links statutory and economic incidence. Germany’s health insurance “add-on premium” was levied fully on employees before 2019, when it was split evenly between employers and employees without changing the nominal rate. The shift aimed at redistributing the economic burden.¹¹ In Finland, to reduce non-wage labor costs, unions and employers agreed to raise employees’ pension contributions by 1.2pp between 2017 and 2020 and raise employees’ unemployment insurance contributions by 0.85pp between 2017 and 2018 while lowering the employer shares by the respective amounts. Our theoretical insights imply that shifting the statutory incidence toward employers at constant rates reduces the overall economic burden (on both workers and firms) whereas shifts toward the employee increase effective rates. Thus, the German reform lowered the effective tax rate, whereas it increased in Finland.

Other reforms illustrate the “traditional” view by simultaneously changing payroll tax rates and statutory incidence. Mexico, for instance, increased employer contributions for old-age support from 5.15% to 13.875% in 2020, raising employer shares with the goal of leaving workers’ income unaffected.¹² Lithuania implemented a reform in 2019 in the opposite direction aimed at reducing the tax burden on labor and stabilizing the tax environment.¹³ Employer rates were decreased from 30.5% to 1.46%, whereas employee rates increased from 9% to 20.96%. Similarly, Romania (a non-OECD country) shifted social contributions from the employer to the employee in 2018. Employee rates increased from 16.5% to 35.0%, while the employer rate fell from 22.75% to 2.25%. The reform was motivated by reducing the administrative burden on employers and improving social contribution collection.¹⁴ While the effective payroll tax rate declined in Lithuania despite the increase in the employee share, the effective tax rate in Romania increased despite a lower nominal rate post-reform.

7 Application of new incidence formulas under tax evasion

While the policy examples above could illustrate economically meaningful effects of statutory incidence changes, each setting complicates clean causal identification. First, simultaneous changes in payroll tax rates and statutory incidence are difficult to disentangle. Second, when only statutory incidence changes, as in Germany, variation is small and resulting effects are unlikely to be precisely estimated. Third, in Finland, statutory incidence changes

¹¹[Link to Federal Ministry of Health](#), accessed: 10/2025.

¹²[Link to Ministry of Finance Mexico](#), accessed: 10/2025

¹³[Link to Ministry of Finance Lithuania](#), accessed: 10/2025

¹⁴[Link to brief by the European Commission](#), accessed: 10/2025

coincide with broader payroll reforms, making attribution challenging. Other empirical applications involve institutional features outside our two-agent framework (e.g., [Hansen et al. \(2026\)](#)). Nevertheless, policies aimed at reducing tax evasion provide a recurring setting for statutory incidence changes and allow us to apply our incidence formulas, as in [Bibler et al. \(2021\)](#). We combine their reduced-form estimates with our evasion-adjusted formulas to show that our framework replicates and refines their findings.

[Bibler et al. \(2021\)](#) study a statutory incidence shift from the supply to the demand side ($d\alpha = 1$), where Airbnb hosts were previously responsible for reporting and remitting taxes. They find that prior to enforcement, at most 24% of Airbnb transactions were taxed. Following the reform—where taxes were automatically withheld at booking—prices received by hosts decreased by 2.4% ($\frac{dp_S}{p_S}$), while prices paid by renters increased by 7.6% ($\frac{dp_D}{p_D}$), at a tax rate of $\tau = 0.1$. While a mechanical shift in statutory incidence toward the demand side ($d\alpha > 0$) would predict opposite comparative statics in a frictionless benchmark, the observed pattern is consistent with low pre-reform compliance (result 3). [Bibler et al. \(2021\)](#) estimate a reduction in nights booked of 3.61%, implying a demand elasticity of -0.475 , and obtain a supply elasticity of at least 1.5 in a bounding exercise. Using the ratio of equations (6) and (7), and assuming that the structural compliance parameter is unaffected by the reform ($d\gamma = 0$), we can instead *point*-identify the supply elasticity:

$$\frac{\frac{dp_S}{p_S}}{\frac{dp_D}{p_D}} = \frac{\epsilon^D}{\epsilon^S} \Rightarrow \epsilon^S = 1.504$$

Finally, using either equation (6) or (7) with $\alpha = 0$, $\tau = 0.1$, and $d\alpha = 1$, we can refine the evasion estimate, obtaining $\gamma \approx 0.8\%$. This estimate is very close to the 0 to 1.6% pre-reform compliance found in [Bibler et al. \(forthcoming\)](#), who provide a complementary methodology to point-identify elasticities and tax evasion following tax enforcement variation.

8 Conclusion

This paper reexamined the classical irrelevance of statutory incidence for ad valorem taxes. For tax incidence research, our results allow for a better decomposition of observed price and quantity effects into a mechanical effect as expected in competitive equilibrium absent any frictions and additional effects due to optimization frictions or evasion opportunities. For policy, our results demonstrate that shifts in the statutory incidence imply effective tax rate changes. These insights can guide our understanding of how statutory incidence of ad valorem taxes affects economic incidence, and are important for building more elaborate theories to explain tax anomalies ([Benzarti, 2025](#)).

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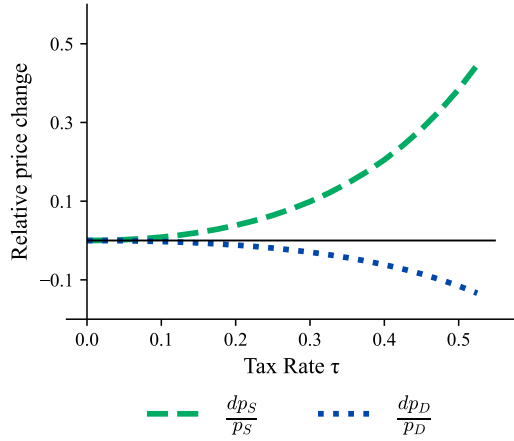
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Figures

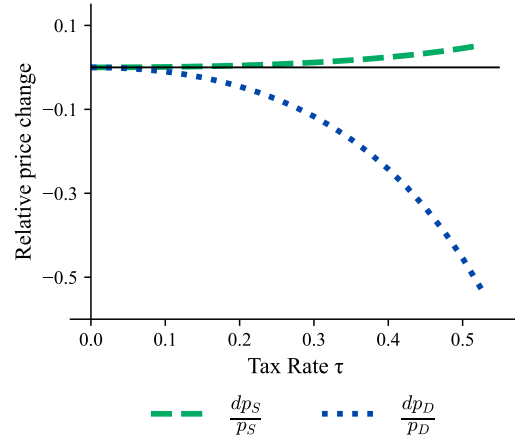
Figure 1: Simulations of the Effect of Tax Policy Changes

I. Effect of shifting the statutory incidence from the supply to the demand side

(a) $\epsilon^S = 0.3, \epsilon^D = -1$

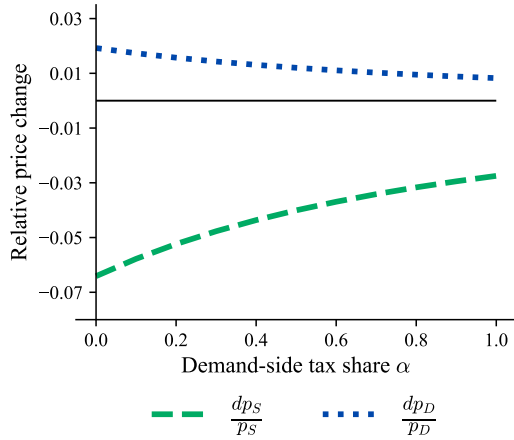


(b) $\epsilon^S = 0.1, \epsilon^D = -0.01$

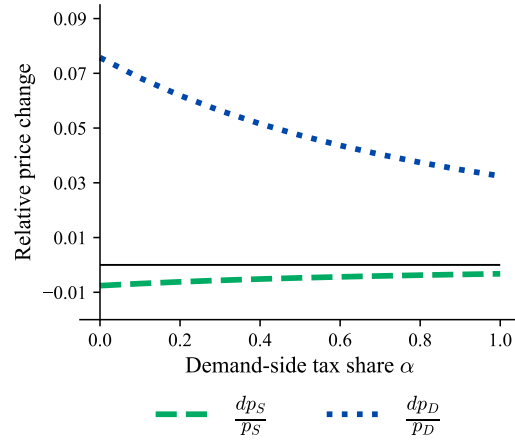


II. Numerical Examples for Effect of a Change in Tax Rate on Relative Price Changes

(c) $\epsilon^S = 0.3, \epsilon^D = -1$



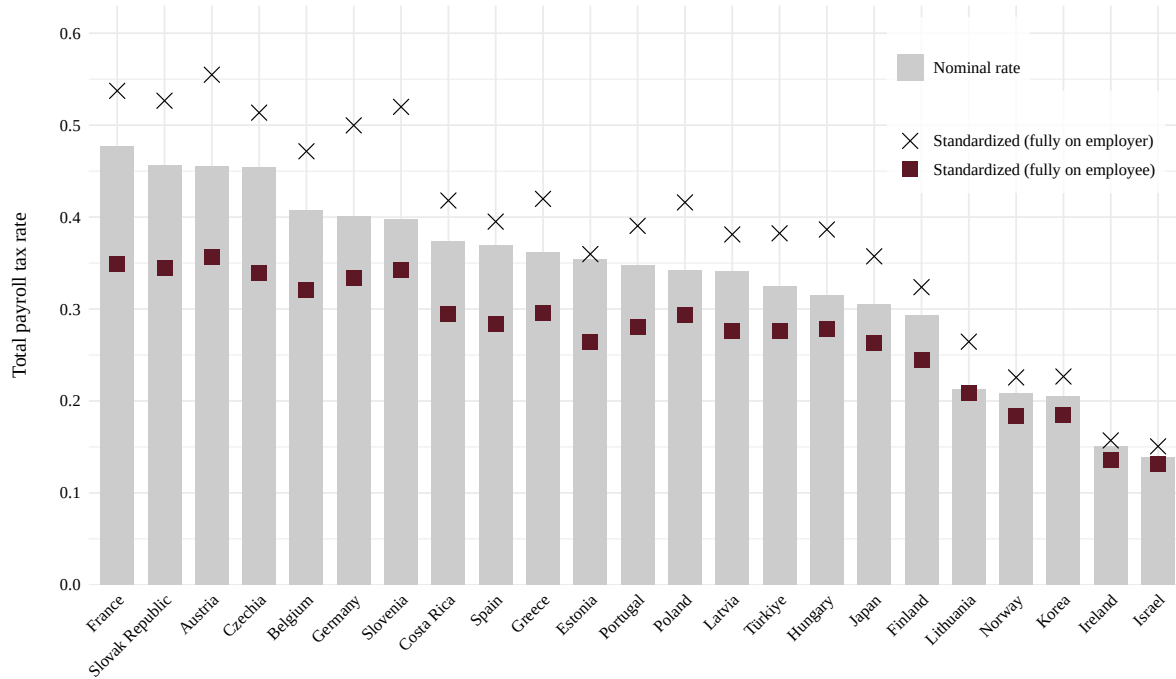
(d) $\epsilon^S = 0.1, \epsilon^D = -0.01$



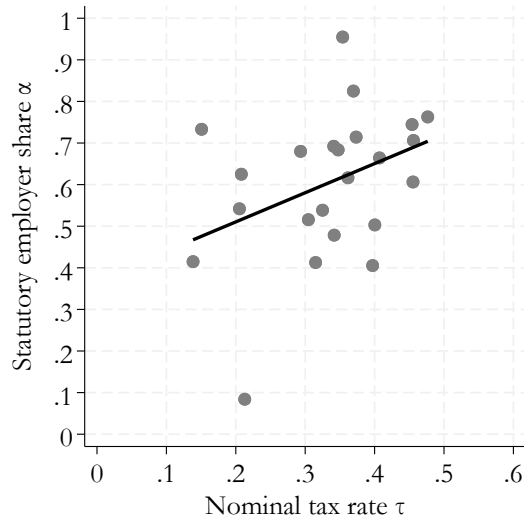
Notes: This figure shows numerical examples of the effect of tax policy changes with elasticities and tax rates calibrated to the payroll tax context. Part I shows numerical examples for the effect of changing the statutory incidence from the supply side ($\alpha = 0$) to the demand side ($\alpha = 1$) on relative price changes. Each panel shows the effect on the relative demand-side price change in blue (dotted line) and on the relative supply-side change in green (dashed line) for different levels of the tax rate τ . Panel (a) corresponds to a supply elasticity of 0.3 and a demand elasticity of -1. Panel (b) corresponds to a supply elasticity of 0.1 and a demand elasticity of -0.01. Part II shows numerical examples for the effect of changing the tax rate from 0.4 to 0.45 on relative price changes at different levels of statutory incidence α . Panel (c) corresponds to a supply elasticity of 0.3 and a demand elasticity of -1. Panel (d) corresponds to a supply elasticity of 0.1 and a demand elasticity of -0.01.

Figure 2: Payroll taxes and statutory incidence across OECD countries

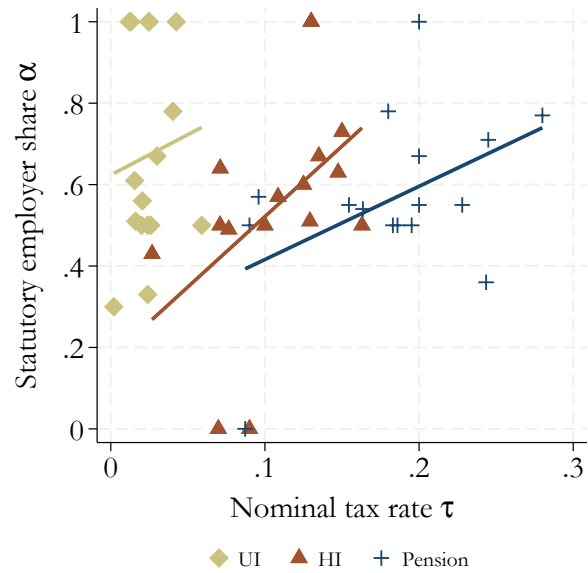
(a) Nominal and standardized payroll tax rates



(b) Correlation between τ and α



(c) Correlation between τ and α by type



Notes: This figure shows payroll tax rates and corresponding statutory incidences for OECD countries in 2024. In Panel (a), each gray bar corresponds to the total nominal tax rate. Each black X corresponds to the standardized tax rate when it falls completely on the employer ($\alpha = 1$). Each red square corresponds to the standardized tax rate when it falls completely on the employee ($\alpha = 0$). Panel (b) plots each country's nominal payroll tax rate against its statutory incidence. Panel (c) shows nominal unemployment (yellow diamonds), health (orange triangles), and pension (blue pluses) insurance tax rates and their respective statutory incidences.

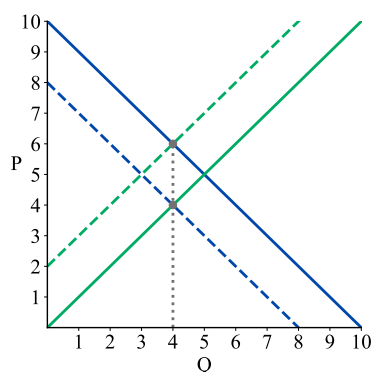
Online Appendix

A Appendix Figures

Figure A.1: (Ir)relevance of statutory incidence for per unit vs. ad valorem taxes

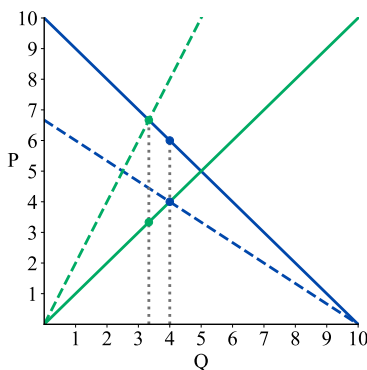
(a) Irrelevance per unit tax

$$t_D = t_S$$



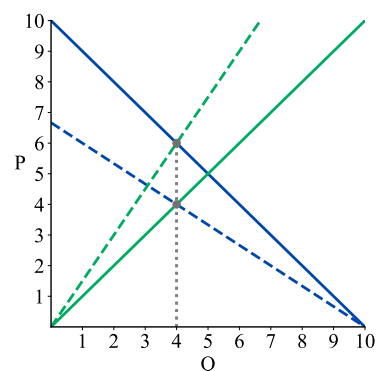
(b) Strong relevance ad valorem tax

$$\tau_D = \tau_S$$



(c) Weak irrelevance ad valorem tax

$$\tau_D = \tau_S / (1 - \tau_S)$$



Notes: This figure illustrates in classic price-quantity diagrams the (ir)relevance of statutory tax incidence for per unit and ad valorem taxes. Downward-sloping, solid blue lines correspond to inverse linear demand curves. Upward-sloping, solid green lines correspond to inverse linear supply curves. Dashed blue and green lines account for taxes when levied on the supply and demand side, respectively. Panel (a) shows that for a per unit tax ($t_D = t_S$), the statutory incidence affects neither equilibrium quantity nor prices. Panel (b) demonstrates the economic relevance of statutory tax incidence for fixed ad valorem tax rates ($\tau_D = \tau_S$). Panel (c) illustrates the case where different ad valorem tax rates on the supply side and the demand side ($\tau_D = \tau_S / (1 - \tau_S)$) lead to the same equilibrium (the case of weak irrelevance).

B Detailed derivations

B.1 Per unit taxes — detailed derivations

B.1.1 Economic incidence of per unit taxes

We consider the case $d\alpha = 0$ and $dt \neq 0$ for arbitrary $\alpha \in [0, 1]$. We start by totally differentiating the equilibrium condition $S(p - (1 - \alpha)t) = D(p + \alpha t)$ and using $p_S = p_D - t$, which results in

$$\begin{aligned}\frac{\partial S}{\partial p_S} [dp - (1 - \alpha)dt] &= \frac{\partial D}{\partial p_D} [dp + \alpha dt] \\ \Leftrightarrow \frac{dp}{dt} \left[\frac{\partial S}{\partial p_S} - \frac{\partial D}{\partial p_D} \right] &= \frac{\partial S}{\partial p_S} (1 - \alpha) + \frac{\partial D}{\partial p_D} \alpha \\ \Leftrightarrow \frac{dp}{dt} &= \frac{(1 - \alpha)\varepsilon^S + \alpha\varepsilon^D - \frac{\partial D}{\partial p_D} \frac{t}{D}}{\varepsilon^S - \varepsilon^D + \frac{\partial D}{\partial p_D} \frac{t}{D}}.\end{aligned}$$

Setting $t = 0, \alpha = 1$, we derive the classical textbook result:

$$\frac{dp}{dt} = \frac{\varepsilon^D}{\varepsilon^S - \varepsilon^D}.$$

However, the effects on p_S and p_D are independent of the statutory incidence: First, consider $p_S = p - (1 - \alpha)t$. Totally differentiating both sides yields:

$$\begin{aligned}dp_S &= dp - (1 - \alpha)dt \\ \Leftrightarrow \frac{dp_S}{dt} &= \frac{\varepsilon^D - \frac{\partial D}{\partial p_D} \frac{t}{D}}{\varepsilon^S - \varepsilon^D + \frac{\partial D}{\partial p_D} \frac{t}{D}}.\end{aligned}$$

Similarly, we totally differentiate $p_D = p_S + t$, yielding:

$$\frac{dp_D}{dt} = \frac{\varepsilon^S}{\varepsilon^S - \varepsilon^D + \frac{\partial D}{\partial p_D} \frac{t}{D}}.$$

Both yield the classical textbook result for $t = 0$.

B.1.2 Effect of simultaneous changes in tax rate and statutory incidence

We start by totally differentiating the equilibrium condition $S(p - (1 - \alpha)t) = D(p + \alpha t)$ and using $p_S = p_D - t$, which results in

$$\begin{aligned} \frac{\partial S}{\partial p_S} [dp - (1 - \alpha)dt + t d\alpha] &= \frac{\partial D}{\partial p_D} [dp + d\alpha t + \alpha dt] \\ \Leftrightarrow \frac{dp}{dt} \left[\frac{\partial S}{\partial p_S} - \frac{\partial D}{\partial p_D} \right] + \frac{d\alpha t}{dt} \left[\frac{\partial S}{\partial p_S} - \frac{\partial D}{\partial p_D} \right] &= \frac{\partial S}{\partial p_S} (1 - \alpha) + \frac{\partial D}{\partial p_D} \alpha \\ \Leftrightarrow \frac{dp}{dt} &= -\frac{d\alpha t}{dt} + \frac{(1 - \alpha)\epsilon^S + \alpha\epsilon^D - \frac{\partial D}{\partial p_D} \frac{t}{D}}{\epsilon^S - \epsilon^D + \frac{\partial D}{\partial p_D} \frac{t}{D}}. \end{aligned}$$

Again, the effects on p_S and p_D are independent of the statutory incidence: First, consider $p_S = p - (1 - \alpha)t$. Totally differentiating both sides yields:

$$\begin{aligned} dp_S &= dp - (1 - \alpha)dt + t d\alpha \\ \Leftrightarrow \frac{dp_S}{dt} &= \frac{\epsilon^D - \frac{\partial D}{\partial p_D} \frac{t}{D}}{\epsilon^S - \epsilon^D + \frac{\partial D}{\partial p_D} \frac{t}{D}}. \end{aligned}$$

Similarly, we totally differentiate $p_D = p_S + t$, yielding:

$$\frac{dp_D}{dt} = \frac{\epsilon^S}{\epsilon^S - \epsilon^D + \frac{\partial D}{\partial p_D} \frac{t}{D}}.$$

B.2 Ad valorem taxes — detailed derivations

B.2.1 Details on the effects of a change in statutory incidence only

Derivation of result 1. We start by totally differentiating the equilibrium condition $S([1 - (1 - \alpha)\tau]p) = D((1 + \alpha\tau)p)$, which results in

$$\begin{aligned} \frac{\partial S}{\partial p_S} [[1 - (1 - \alpha)\tau]dp + \tau p d\alpha] &= \frac{\partial D}{\partial p_D} [(1 + \alpha\tau)dp + \tau p d\alpha] \\ \Leftrightarrow \frac{\partial S}{\partial p_S} \left[[1 - (1 - \alpha)\tau] \frac{dp}{d\alpha} + \tau p \right] &= \frac{\partial D}{\partial p_D} \left[(1 + \alpha\tau) \frac{dp}{d\alpha} + \tau p \right] \\ \Leftrightarrow \frac{dp}{d\alpha} \left[[1 - (1 - \alpha)\tau] \frac{\partial S}{\partial p_S} - [1 + \alpha\tau] \frac{\partial D}{\partial p_D} \right] &= -\tau \left[\frac{\partial S}{\partial p_S} p - \frac{\partial D}{\partial p_D} p \right]. \end{aligned}$$

Using the fact that $S = D$ in equilibrium, we can write

$$\frac{dp}{d\alpha} \left[[1 - (1 - \alpha)\tau] \frac{\partial S}{\partial p_S} \frac{p_S}{S} \frac{1}{p_S} - [1 + \alpha\tau] \frac{\partial D}{\partial p_D} \frac{p_D}{D} \frac{1}{p_D} \right] = -\tau \left[\frac{\partial S}{\partial p_S} \frac{p_S}{S} \frac{1}{p_S} p - \frac{\partial D}{\partial p_D} \frac{p_D}{D} \frac{1}{p_D} p \right],$$

which given the definition of ε^S and ε^D simplifies to

$$\frac{dp}{d\alpha} \left[\frac{1 - (1 - \alpha)\tau}{p_S} \varepsilon^S - \frac{1 + \alpha\tau}{p_D} \varepsilon^D \right] = -\tau \left[\varepsilon^S \frac{p}{p_S} - \varepsilon^D \frac{p}{p_D} \right].$$

Using the relationship between p_D, p_S and p , this further simplifies to

$$\frac{dp}{d\alpha} \frac{1}{p} (\varepsilon^S - \varepsilon^D) = -\tau \left[\frac{\varepsilon^S}{1 - (1 - \alpha)\tau} - \frac{\varepsilon^D}{1 + \alpha\tau} \right].$$

Re-arranging gives result 1. □

Derivation of result 2. First, let us consider $p_S = [1 - (1 - \alpha)\tau]p$. Totally differentiating and holding τ constant, we obtain

$$\begin{aligned} dp_S &= [1 - (1 - \alpha)\tau]dp + \tau p d\alpha \\ \Leftrightarrow \frac{dp_S}{p_S} &= \frac{1 - (1 - \alpha)\tau}{p_S} dp + \tau \frac{p}{p_S} d\alpha \\ &= \frac{dp}{p} + \tau d\alpha \frac{1}{1 - (1 - \alpha)\tau}. \end{aligned}$$

Plugging in (3) for $\frac{dp}{p}$, we obtain

$$\begin{aligned} \frac{dp_S}{p_S} &= \tau d\alpha \left[\frac{-\frac{\varepsilon^S}{1 - (1 - \alpha)\tau} + \frac{\varepsilon^D}{1 + \alpha\tau}}{\varepsilon^S - \varepsilon^D} + \frac{1}{1 - (1 - \alpha)\tau} \right] \\ &= \tau d\alpha \left[\frac{-\varepsilon^S + \frac{1 - (1 - \alpha)\tau}{1 + \alpha\tau} \varepsilon^D + \varepsilon^S - \varepsilon^D}{[1 - (1 - \alpha)\tau](\varepsilon^S - \varepsilon^D)} \right]. \end{aligned}$$

Further simplifying gives (4).

Second, let us consider $p_D = (1 + \alpha\tau)p$. Totally differentiating and holding τ constant, we

obtain

$$\begin{aligned}
dp_D &= (1 + \alpha\tau)dp + \tau p d\alpha \\
\Leftrightarrow \frac{dp_D}{p_D} &= \frac{1 + \alpha\tau}{p_D} dp + \tau \frac{p}{p_D} d\alpha \\
&= \frac{dp}{p} + \frac{1}{1 + \alpha\tau} \tau d\alpha \\
&= -\tau d\alpha \left[\frac{\frac{\epsilon^S}{1 - (1 - \alpha)\tau} - \frac{\epsilon^D}{1 + \alpha\tau}}{\epsilon^S - \epsilon^D} - \frac{1}{1 + \alpha\tau} \right] \\
&= -\tau d\alpha \left[\frac{\frac{1 + \alpha\tau}{1 - (1 - \alpha)\tau} \epsilon^S - \epsilon^D - \epsilon^S + \epsilon^D}{(1 + \alpha\tau)(\epsilon^S - \epsilon^D)} \right].
\end{aligned}$$

Further simplifying gives (5). □

Extended proof of corollary 2. This proof uses a perturbation argument. Assume $\alpha \in [0, 1)$ and consider a reform that shifts the incidence towards the demand side, i.e., $d\alpha > 0$. Equations (4) and (5) show that $\frac{dp_S}{p_S} \geq 0$ and $\frac{dp_D}{p_D} \leq 0$. Given upward sloping supply and downward sloping demand curves, this implies that the equilibrium quantity is increasing in α .

More formally, let us first totally differentiate $D((1 + \alpha\tau)p)$, holding τ fixed and using (1),

$$\begin{aligned}
dD^* &= \frac{\partial D}{\partial p_D} [(1 + \alpha\tau)dp + \tau p d\alpha] \\
&\propto \frac{\partial D}{\partial p_D} \left[(1 + \alpha\tau) \frac{dp}{p} + \tau d\alpha \right] \\
&= \frac{\partial D}{\partial p_D} \left[\tau \alpha \left(1 - \frac{\frac{1 + \alpha\tau}{1 - (1 - \alpha)\tau} \epsilon^S - \epsilon^D}{\epsilon^S - \epsilon^D} \right) \right] \\
&= d\alpha \left(-\frac{\partial D}{\partial p_D} \right) \frac{\tau^2}{1 - (1 - \alpha)\tau} \frac{-\epsilon^S}{\epsilon^S - \epsilon^D} \\
&= d\alpha \frac{\tau}{(1 + \alpha\tau)} \frac{\tau}{[1 - (1 - \alpha)\tau]} \frac{Q^* \epsilon^S * (-\epsilon^D)}{p \frac{\epsilon^S - \epsilon^D}{\epsilon^S - \epsilon^D}} \\
&\geq 0.
\end{aligned}$$

Next, we consider $S([1 - (1 - \alpha)\tau]p)$, which gives

$$\begin{aligned}
dS^* &= \frac{\partial S}{\partial p_S} [[1 - (1 - \alpha)\tau]dp + \tau p d\alpha] \\
&\propto \frac{\partial S}{\partial p_S} \left[[1 - (1 - \alpha)\tau] \frac{dp}{p} + \tau d\alpha \right] \\
&= \frac{\partial S}{\partial p_S} \left[\tau d\alpha \left(1 - \frac{\varepsilon^S - \frac{1-(1-\alpha)\tau}{1+\alpha\tau} \varepsilon^D}{\varepsilon^S - \varepsilon^D} \right) \right] \\
&= \tau^2 d\alpha \frac{\partial S}{\partial p_S} \frac{1}{1 + \alpha\tau} \frac{-\varepsilon^D}{\varepsilon^S - \varepsilon^D} \\
&= d\alpha \frac{\tau}{(1 + \alpha\tau)} \frac{\tau}{[1 - (1 - \alpha)\tau]} \frac{Q^* \varepsilon^S * (-\varepsilon^D)}{p \varepsilon^S - \varepsilon^D} \\
&\geq 0.
\end{aligned}$$

□

B.2.2 Details on change in tax rate only

Consider a tax-reform that changes the tax rate by $d\tau \neq 0$ while keeping the statutory incidence unchanged, i.e., $d\alpha = 0$. Then, the tax-exclusive price changes by

$$\frac{dp}{d\tau} \frac{1}{p} = \frac{\frac{1-\alpha}{1-(1-\alpha)\tau} \varepsilon^S + \frac{\alpha}{1+\alpha\tau} \varepsilon^D}{(\varepsilon^S - \varepsilon^D)}. \quad (11)$$

This follows from totally differentiating the equilibrium condition $S([1 - (1 - \alpha)\tau]p) = D((1 + \alpha\tau)p)$ with $d\tau \neq 0$ and $d\alpha = 0$.

Derivation of result 4. We start by totally differentiating the equilibrium condition $S([1 - (1 - \alpha)\tau]p) = D((1 + \alpha\tau)p)$ with $d\alpha = 0$ & $d\tau \neq 0$, allowing for endogenous response in p (market clearing), which results in

$$\begin{aligned}
&\frac{\partial S}{\partial p_S} [1 - (1 - \alpha)\tau]dp - (1 - \alpha)p d\tau = \frac{\partial D}{\partial p_D} [(1 + \alpha\tau)dp + \alpha p d\tau] \\
\Leftrightarrow \frac{\partial S}{\partial p_S} \left[1 - (1 - \alpha)\tau \right] \frac{dp}{d\tau} - (1 - \alpha)p &= \frac{\partial D}{\partial p_D} \left[(1 + \alpha\tau) \frac{dp}{d\tau} + \alpha p \right] \\
\Leftrightarrow \frac{\partial S}{\partial p_S} \left[\frac{p_S}{p} \frac{dp}{d\tau} - (1 - \alpha)p \right] &= \frac{\partial D}{\partial p_D} \left[\frac{p_D}{p} \frac{dp}{d\tau} + \alpha p \right] \\
\Leftrightarrow \frac{dp}{p} \frac{1}{d\tau} \left[\frac{\partial S}{\partial p_S} p_S - \frac{\partial D}{\partial p_D} p_D \right] &= (1 - \alpha) \frac{\partial S}{\partial p_S} p + \alpha \frac{\partial D}{\partial p_D} p.
\end{aligned}$$

Using the fact that $S = D$ in equilibrium, and the definition of ε^S and ε^D simplifies to

$$\frac{dp}{p} \frac{1}{d\tau} (\varepsilon^S - \varepsilon^D) = (1 - \alpha) \varepsilon^S \frac{p}{p_S} + \alpha \varepsilon^D \frac{p}{p_D}.$$

Using the relationship between p_D , p_S and p , this further simplifies to

$$\frac{dp}{d\tau} \frac{1}{p} = \frac{\frac{1-\alpha}{1-(1-\alpha)\tau} \varepsilon^S + \frac{\alpha}{1+\alpha\tau} \varepsilon^D}{(\varepsilon^S - \varepsilon^D)}.$$

□

Derivation of result 4. First, we derive the tax-inclusive prices. Let us consider $p_S = [1 - (1 - \alpha)\tau]p$. Totally differentiating and holding α constant, we obtain

$$\begin{aligned} dp_S &= [1 - (1 - \alpha)\tau]dp - (1 - \alpha)p d\tau \\ \Leftrightarrow \frac{dp_S}{d\tau} \frac{1}{p_S} &= \frac{1 - (1 - \alpha)\tau}{p_S} \frac{dp}{d\tau} - (1 - \alpha) \frac{p}{p_S} \\ &= \frac{dp}{d\tau} \frac{1}{p} - \frac{1 - \alpha}{1 - (1 - \alpha)\tau} \\ &= \frac{\frac{1-\alpha}{1-(1-\alpha)\tau} \varepsilon^S + \frac{\alpha}{1+\alpha\tau} \varepsilon^D}{(\varepsilon^S - \varepsilon^D)} - \frac{1 - \alpha}{1 - (1 - \alpha)\tau} \\ &= \frac{1}{[1 - (1 - \alpha)\tau] (1 + \alpha\tau)} \frac{\varepsilon^D}{(\varepsilon^S - \varepsilon^D)}. \end{aligned}$$

Second, we consider $p_D = [1 + \alpha\tau]p$. Totally differentiating and holding α constant, we obtain

$$\begin{aligned} dp_D &= [1 + \alpha\tau]dp + \alpha p d\tau \\ \Leftrightarrow \frac{dp_D}{d\tau} \frac{1}{p_D} &= \frac{1 + \alpha\tau}{p_D} \frac{dp}{d\tau} + \alpha \frac{p}{p_D} \\ &= \frac{dp}{d\tau} \frac{1}{p} + \frac{\alpha}{1 + \alpha\tau} \\ &= \frac{\frac{1-\alpha}{1-(1-\alpha)\tau} \varepsilon^S + \frac{\alpha}{1+\alpha\tau} \varepsilon^D}{(\varepsilon^S - \varepsilon^D)} + \frac{\alpha}{1 + \alpha\tau} \\ &= \frac{1}{[1 - (1 - \alpha)\tau] (1 + \alpha\tau)} \frac{\varepsilon^S}{(\varepsilon^S - \varepsilon^D)}. \end{aligned}$$

□

B.2.3 Details on simultaneous changes in tax rate and statutory incidence of ad valorem taxes

Derivation. We consider the case $d\alpha \neq 0$ & $d\tau \neq 0$. We start by totally differentiating the equilibrium condition $S([1 - (1 - \alpha)\tau]p) = D((1 + \alpha\tau)p)$, allowing for endogenous response in p (market clearing).

$$\begin{aligned}
\frac{\partial S}{\partial p_S} [[1 - (1 - \alpha)\tau]dp - (1 - \alpha)p d\tau + \tau p d\alpha] &= \frac{\partial D}{\partial p_D} [(1 + \alpha\tau)dp + \alpha p d\tau + \tau p d\alpha] \\
\Leftrightarrow \frac{dp}{p} \left[\frac{\partial S}{\partial p_S} [1 - (1 - \alpha)\tau] - \frac{\partial D}{\partial p_D} (1 + \alpha\tau) \right] &= \left[\frac{\partial D}{\partial p_D} \alpha + \frac{\partial S}{\partial p_S} (1 - \alpha) \right] d\tau + \left[\frac{\partial D}{\partial p_D} - \frac{\partial S}{\partial p_S} \right] \tau d\alpha \\
\Leftrightarrow \frac{dp}{p} \left[\frac{\partial S}{\partial p_S} \frac{p_S}{p} - \frac{\partial D}{\partial p_D} \frac{p_D}{p} \right] &= \left[\frac{\partial D}{\partial p_D} \alpha + \frac{\partial S}{\partial p_S} (1 - \alpha) \right] d\tau + \left[\frac{\partial D}{\partial p_D} - \frac{\partial S}{\partial p_S} \right] \tau d\alpha \\
\Leftrightarrow \frac{dp}{p} [\varepsilon^S - \varepsilon^D] &= \left[\frac{\partial D}{\partial p_D} \frac{p}{D} \alpha + \frac{\partial S}{\partial p_S} \frac{p}{S} (1 - \alpha) \right] d\tau + \left[\frac{\partial D}{\partial p_D} \frac{p}{D} - \frac{\partial S}{\partial p_S} \frac{p}{S} \right] \tau d\alpha \\
\Leftrightarrow \frac{dp}{p} [\varepsilon^S - \varepsilon^D] &= \left[\frac{\alpha}{1 + \alpha\tau} \varepsilon^D + \frac{1 - \alpha}{1 + (1 - \alpha)\tau} \varepsilon^S \right] d\tau + \\
&\quad \left[\frac{1}{1 + \alpha\tau} \varepsilon^D + \frac{1}{1 + (1 - \alpha)\tau} \varepsilon^S \right] \tau d\alpha \\
\frac{dp}{p} &= \underbrace{\frac{\frac{\alpha}{1 + \alpha\tau} \varepsilon^D + \frac{1 - \alpha}{1 + (1 - \alpha)\tau} \varepsilon^S}{\varepsilon^S - \varepsilon^D} d\tau}_{\text{Effect when changing } \tau \text{ only}} - \underbrace{\frac{\frac{1}{1 + (1 - \alpha)\tau} \varepsilon^S - \frac{1}{1 + \alpha\tau} \varepsilon^D}{\varepsilon^S - \varepsilon^D} \tau d\alpha}_{\text{Effect when changing } \alpha \text{ only}}
\end{aligned}$$

□

Derivation of tax-inclusive prices. Next, we derive the tax-inclusive prices. First, let us consider $p_S = [1 - (1 - \alpha)\tau]p$. Totally differentiating, we obtain

$$\begin{aligned}
dp_S &= [1 - (1 - \alpha)\tau]dp - (1 - \alpha)p d\tau + \tau p d\alpha \\
\Leftrightarrow \frac{dp_S}{p_S} &= \frac{dp}{p} - \frac{1 - \alpha}{[1 - (1 - \alpha)\tau]} d\tau + \frac{\tau}{1 - (1 - \alpha)\tau} d\alpha \\
\Leftrightarrow \frac{dp_S}{p_S} &= \frac{1}{[1 - (1 - \alpha)\tau] (1 + \alpha\tau)} \frac{\varepsilon^D}{(\varepsilon^S - \varepsilon^D)} d\tau + \frac{\tau^2}{(1 + \alpha\tau)[1 - (1 - \alpha)\tau]} \frac{-\varepsilon^D}{(\varepsilon^S - \varepsilon^D)} d\alpha
\end{aligned}$$

Second, we consider $p_D = [1 + \alpha\tau]p$. Totally differentiating, we obtain

$$\begin{aligned}
 dp_D &= [1 + \alpha\tau]dp + \alpha p d\tau + \tau p d\alpha \\
 \Leftrightarrow \frac{dp_D}{p_D} &= \frac{dp}{p} + \frac{\alpha}{[1 + \alpha\tau]}d\tau + \frac{\tau}{1 + \alpha\tau}d\alpha \\
 \Leftrightarrow \frac{dp_D}{p_D} &= \frac{1}{[1 - (1 - \alpha)\tau](1 + \alpha\tau)} \frac{\varepsilon^S}{(\varepsilon^S - \varepsilon^D)} d\tau - \frac{\tau^2}{(1 + \alpha\tau)[1 - (1 - \alpha)\tau]} \frac{-\varepsilon^S}{(\varepsilon^S - \varepsilon^D)} d\alpha
 \end{aligned}$$

□

B.3 Closed form illustration of ad valorem relevance result

The previous derivations suggest post-tax wages for employees, $\tilde{w}$, are higher when the statutory incidence falls on the employer, while the employer's post-tax labor cost per unit, $\hat{w}$, is lower simultaneously. We illustrate this here for an economy with a unit mass of identical households and a representative firm, assuming simple functional forms for the households' utility and the firm's production function.

Optimization problems. Specifically, assume households solve the following optimization problem.

$$\max_l \tilde{w}l - \frac{l^{1+\frac{1}{\sigma}}}{1+\frac{1}{\sigma}}$$

This gives an aggregate labor supply of $L^S(\tilde{w}) = \tilde{w}^\sigma$. Further, assume the representative firm demands labor according to the following profit maximization problem.

$$\max_L L^{1-\beta} - \hat{w}L$$

This results in an aggregate labor demand of $L^D(\hat{w}) = (1-\beta)^{\frac{1}{\beta}} \hat{w}^{-\frac{1}{\beta}}$.

Statutory incidence on the employee. We first consider the case of $\alpha = 0$. Then, $\tilde{w} = (1-\tau)w$ and $\hat{w} = w$. In equilibrium, the wage rate w adjusts such that $L^D = L^S$. Solving for this, we obtain

$$\begin{aligned} w &= (1-\beta)^{\frac{1}{1+\sigma\beta}} (1-\tau)^{-\frac{\sigma\beta}{1+\sigma\beta}} \\ \tilde{w} &= (1-\beta)^{\frac{1}{1+\sigma\beta}} (1-\tau)^{\frac{1}{1+\sigma\beta}} \\ \hat{w} &= w \end{aligned}$$

Statutory incidence on the employer. Next, we consider the case of $\alpha = 1$. Then, $\tilde{w} = w$ and $\hat{w} = (1+\tau)w$. Again, the wage rate w adjusts such that $L^D = L^S$. Thus, we obtain

$$\begin{aligned} w &= (1-\beta)^{\frac{1}{1+\sigma\beta}} (1+\tau)^{-\frac{1}{1+\sigma\beta}} \\ \tilde{w} &= w \\ \hat{w} &= (1-\beta)^{\frac{1}{1+\sigma\beta}} (1+\tau)^{\frac{\sigma\beta}{1+\sigma\beta}} \end{aligned}$$

Comparison of statutory incidence regimes. Given the results above, we confirm that $\tilde{w}_{\alpha=0} < \tilde{w}_{\alpha=1}$ and $\hat{w}_{\alpha=0} > \hat{w}_{\alpha=1}$ for any $\tau > 0$ independent of σ and β . That is, workers earn more post-taxes per unit of labor supplied when the statutory incidence falls on the employer. Employers, on the other hand, face lower post-tax labor costs per unit when the statutory

incidence falls on the employee. Moreover, since the amount of labor employed in equilibrium is increasing in $\tilde{w}$, employment and output are higher when the nominal incidence falls on the employer.

B.4 Ad valorem taxes extensions — detailed derivations

B.4.1 Details on change in statutory incidence of ad valorem subsidies

Derivation of corollary 4. We start by totally differentiating the equilibrium condition $S([1 + (1 - \alpha)\tau]p) = D((1 - \alpha\tau)p)$, which results in

$$\begin{aligned} \frac{\partial S}{\partial p_S} [[1 + (1 - \alpha)\tau]dp - \tau p d\alpha] &= \frac{\partial D}{\partial p_D} [(1 - \alpha\tau)dp - \tau p d\alpha] \\ \Leftrightarrow \frac{\partial S}{\partial p_S} \left[[1 + (1 - \alpha)\tau] \frac{dp}{d\alpha} - \tau p \right] &= \frac{\partial D}{\partial p_D} \left[(1 - \alpha\tau) \frac{dp}{d\alpha} - \tau p \right] \\ \Leftrightarrow \frac{dp}{d\alpha} \left[[1 + (1 - \alpha)\tau] \frac{\partial S}{\partial p_S} - (1 - \alpha\tau) \frac{\partial D}{\partial p_D} \right] &= \tau \left[\frac{\partial S}{\partial p_S} p - \frac{\partial D}{\partial p_D} p \right]. \end{aligned}$$

Using the fact that $S = D$ in equilibrium, we can write

$$\frac{dp}{d\alpha} \left[[1 + (1 - \alpha)\tau] \frac{\partial S}{\partial p_S} \frac{p_S}{S} \frac{1}{p_S} - (1 - \alpha\tau) \frac{\partial D}{\partial p_D} \frac{p_D}{D} \frac{1}{p_D} \right] = \tau \left[\frac{\partial S}{\partial p_S} \frac{p_S}{S} \frac{1}{p_S} p - \frac{\partial D}{\partial p_D} \frac{p_D}{D} \frac{1}{p_D} p \right],$$

which given the definition of ε^S and ε^D simplifies to

$$\frac{dp}{d\alpha} \left[\frac{1 + (1 - \alpha)\tau}{p_S} \varepsilon^S - \frac{1 - \alpha\tau}{p_D} \varepsilon^D \right] = \tau \left[\varepsilon^S \frac{p}{p_S} - \varepsilon^D \frac{p}{p_D} \right].$$

Using the relationship between p_D , p_S and p , this further simplifies to

$$\begin{aligned} \frac{dp}{d\alpha} \frac{1}{p} (\varepsilon^S - \varepsilon^D) &= \tau \left[\frac{\varepsilon^S}{1 + (1 - \alpha)\tau} - \frac{\varepsilon^D}{1 - \alpha\tau} \right] \\ \Leftrightarrow \frac{dp}{p} &= d\alpha \frac{\tau \left[\frac{\varepsilon^S}{1 + (1 - \alpha)\tau} - \frac{\varepsilon^D}{1 - \alpha\tau} \right]}{(\varepsilon^S - \varepsilon^D)}. \end{aligned}$$

Next, let us consider $p_S = [1 + (1 - \alpha)\tau]p$. Totally differentiating and holding τ constant,

we obtain

$$\begin{aligned}
dp_S &= [1 + (1 - \alpha)\tau]dp - \tau p d\alpha \\
\Leftrightarrow \frac{dp_S}{p_S} &= \frac{1 + (1 - \alpha)\tau}{p_S} dp - \tau \frac{p}{p_S} d\alpha \\
&= \frac{dp}{p} - \tau d\alpha \frac{1}{1 + (1 - \alpha)\tau}.
\end{aligned}$$

Plugging in for $\frac{dp}{p}$, we obtain

$$\begin{aligned}
\frac{dp_S}{p_S} &= \tau d\alpha \left[\frac{\frac{\varepsilon^S}{1 + (1 - \alpha)\tau} - \frac{\varepsilon^D}{1 - \alpha\tau}}{\varepsilon^S - \varepsilon^D} - \frac{1}{1 + (1 - \alpha)\tau} \right] \\
&= \tau d\alpha \left[\frac{\varepsilon^S - \frac{1 + (1 - \alpha)\tau}{1 - \alpha\tau} \varepsilon^D - \varepsilon^S + \varepsilon^D}{[1 + (1 - \alpha)\tau](\varepsilon^S - \varepsilon^D)} \right].
\end{aligned}$$

Further simplifying gives

$$\frac{dp_S}{p_S} = s d\alpha \frac{s}{(1 - \alpha s)[1 + (1 - \alpha)s]} \frac{-\varepsilon^D}{\varepsilon^S - \varepsilon^D} \quad (12)$$

.

Next, let us consider $p_D = (1 - \alpha\tau)p$. Totally differentiating and holding τ constant, we obtain

$$\begin{aligned}
dp_D &= (1 - \alpha\tau)dp - \tau p d\alpha \\
\Leftrightarrow \frac{dp_D}{p_D} &= \frac{1 - \alpha\tau}{p_D} dp - \tau \frac{p}{p_D} d\alpha \\
&= \frac{dp}{p} - \frac{1}{1 - \alpha\tau} \tau d\alpha \\
&= \tau d\alpha \left[\frac{\frac{\varepsilon^S}{1 + (1 - \alpha)\tau} - \frac{\varepsilon^D}{1 - \alpha\tau}}{\varepsilon^S - \varepsilon^D} - \frac{1}{1 - \alpha\tau} \right] \\
&= \tau d\alpha \left[\frac{\frac{1 - \alpha\tau}{1 + (1 - \alpha)\tau} \varepsilon^S - \varepsilon^D - \varepsilon^S + \varepsilon^D}{(1 - \alpha\tau)(\varepsilon^S - \varepsilon^D)} \right].
\end{aligned}$$

Further simplifying gives

$$\frac{dp_D}{p_D} = s d\alpha \frac{s}{(1 - \alpha s)[1 + (1 - \alpha)s]} \frac{-\varepsilon^S}{\varepsilon^S - \varepsilon^D}. \quad (13)$$

□

B.4.2 Details on change in statutory incidence of ad valorem taxes with imperfect salience

Add the behavioral parameter $\theta \in [0, 1]$ in the expression of (perceived) tax-inclusive prices:

$$\begin{aligned} p_D &= (1 + \alpha\theta\tau)p \\ p_S &= [1 - (1 - \alpha)\tau]p. \end{aligned}$$

Note that θ is isomorphic to alternative interpretations such as a lower compliance rate on the consumer side (e.g., motivated by the evidence in [Kopczuk et al., 2016](#), for diesel taxes). In this specific example, tax revenue would still be $R = \tau pQ = \tau pD$.

Result 5 characterizes expected price changes in the presence of this demand side friction. The effect of changing the statutory incidence continues to depend on the relative elasticities ($\varepsilon^S, \varepsilon^D$), the initial levels of the statutory incidence (α), and the tax rate (τ). In addition, the initial degree of salience (θ) and possible changes in salience ($d\theta$) after shifting the incidence influence the change in prices.

Result 5 (Tax-inclusive price changes due to statutory incidence shifts in the presence of demand-side frictions). *Consider a tax-reform that shifts the statutory incidence by $d\alpha$ without altering the ad valorem tax rate, i.e., $d\tau = 0$. Then, the tax-inclusive prices change by*

$$\frac{dp_S}{p_S} = \tau d\alpha \frac{\theta\tau + 1 - \theta}{(1 + \alpha\theta\tau)[1 - (1 - \alpha)\tau]} \frac{-\varepsilon^D}{\varepsilon^S - \varepsilon^D} + \tau d\theta \frac{\alpha}{(1 + \alpha\theta\tau)} \frac{\varepsilon^D}{\varepsilon^S - \varepsilon^D} \quad (14)$$

$$\frac{dp_D}{p_D} = \tau d\alpha \frac{\theta\tau + 1 - \theta}{(1 + \alpha\theta\tau)[1 - (1 - \alpha)\tau]} \frac{-\varepsilon^S}{\varepsilon^S - \varepsilon^D} + \tau d\theta \frac{\alpha}{(1 + \alpha\theta\tau)} \frac{\varepsilon^S}{\varepsilon^S - \varepsilon^D}. \quad (15)$$

Note that setting $\theta = 1$ and $d\theta = 0$ recovers the frictionless benchmark (result 2). How is the modeled demand-side salience affecting the price changes? Note that $\frac{\theta\tau + 1 - \theta}{(1 + \alpha\theta\tau)[1 - (1 - \alpha)\tau]}$ is decreasing in θ . Suppose for simplicity that $d\theta = 0$ (i.e., view θ as a policy-invariant, structural parameter). Then, price changes in response to statutory incidence shifts ($d\alpha \neq 0$) are stronger the less salient the tax, i.e., the smaller θ . This matters for empirical evaluations of changes in statutory incidence since imperfect salience (e.g. [Chetty et al., 2009](#)) will amplify the expected price changes.

Derivation. We start by totally differentiating the equilibrium condition $S([1 - (1 - \alpha)\tau]p) =$

$D((1 + \alpha\theta\tau)p)$, which results in

$$\begin{aligned} \frac{\partial S}{\partial p_S} [[1 - (1 - \alpha)\tau]dp + \tau p d\alpha] &= \frac{\partial D}{\partial p_D} [(1 + \alpha\theta\tau)dp + \theta\tau p d\alpha + \alpha\tau p d\theta] \\ \Leftrightarrow \frac{\partial S}{\partial p_S} \left[[1 - (1 - \alpha)\tau] \frac{dp}{d\alpha} + \tau p \right] &= \frac{\partial D}{\partial p_D} \left[(1 + \alpha\theta\tau) \frac{dp}{d\alpha} + \theta\tau p + \alpha\tau p \frac{d\theta}{d\alpha} \right] \\ \Leftrightarrow \frac{dp}{d\alpha} \left[[1 - (1 - \alpha)\tau] \frac{\partial S}{\partial p_S} - [1 + \alpha\theta\tau] \frac{\partial D}{\partial p_D} \right] &= -\tau \left[\frac{\partial S}{\partial p_S} p - \frac{\partial D}{\partial p_D} p \left(\theta + \alpha \frac{d\theta}{d\alpha} \right) \right]. \end{aligned}$$

Using the fact that $S = D$ in equilibrium, we can write

$$\frac{dp}{d\alpha} \left[[1 - (1 - \alpha)\tau] \frac{\partial S}{\partial p_S} \frac{p_S}{S} \frac{1}{p_S} - [1 + \alpha\theta\tau] \frac{\partial D}{\partial p_D} \frac{p_D}{D} \frac{1}{p_D} \right] = -\tau \left[\frac{\partial S}{\partial p_S} \frac{p_S}{S} \frac{1}{p_S} p - \frac{\partial D}{\partial p_D} \frac{p_D}{D} \frac{1}{p_D} p \left(\theta + \alpha \frac{d\theta}{d\alpha} \right) \right],$$

which given the definition of ε^S and ε^D simplifies to

$$\frac{dp}{d\alpha} \left[\frac{1 - (1 - \alpha)\tau}{p_S} \varepsilon^S - \frac{1 + \alpha\theta\tau}{p_D} \varepsilon^D \right] = -\tau \left[\varepsilon^S \frac{p}{p_S} - \varepsilon^D \frac{p}{p_D} \left(\theta + \alpha \frac{d\theta}{d\alpha} \right) \right].$$

Using the relationship between p_D, p_S and p , this further simplifies to

$$\begin{aligned} \frac{dp}{d\alpha} \frac{1}{p} (\varepsilon^S - \varepsilon^D) &= -\tau \left[\frac{\varepsilon^S}{1 - (1 - \alpha)\tau} - \frac{\varepsilon^D}{1 + \alpha\theta\tau} \left(\theta + \alpha \frac{d\theta}{d\alpha} \right) \right] \\ \Leftrightarrow \frac{dp}{p} &= d\alpha \frac{-\tau \left[\frac{\varepsilon^S}{1 - (1 - \alpha)\tau} - \frac{\varepsilon^D}{1 + \alpha\theta\tau} \left(\theta + \alpha \frac{d\theta}{d\alpha} \right) \right]}{(\varepsilon^S - \varepsilon^D)}. \end{aligned}$$

Next, let us consider $p_S = [1 - (1 - \alpha)\tau]p$. Totally differentiating and holding τ constant, we obtain

$$\begin{aligned} dp_S &= [1 - (1 - \alpha)\tau]dp + \tau p d\alpha \\ \Leftrightarrow \frac{dp_S}{p_S} &= \frac{1 - (1 - \alpha)\tau}{p_S} dp + \tau \frac{p}{p_S} d\alpha \\ &= \frac{dp}{p} + \tau d\alpha \frac{1}{1 - (1 - \alpha)\tau}. \end{aligned}$$

Plugging in for $\frac{dp}{p}$, we obtain

$$\begin{aligned}\frac{dp_S}{d\alpha} \frac{1}{p_S} &= \tau \left[\frac{-\frac{\varepsilon^S}{1-(1-\alpha)\tau} + \frac{\varepsilon^D}{1+\alpha\theta\tau} \left(\theta + \alpha \frac{d\theta}{d\alpha} \right)}{\varepsilon^S - \varepsilon^D} + \frac{1}{1 - (1-\alpha)\tau} \right] \\ &= \tau \left[\frac{-\varepsilon^S + \frac{1-(1-\alpha)\tau}{1+\alpha\theta\tau} \varepsilon^D \left(\theta + \alpha \frac{d\theta}{d\alpha} \right) + \varepsilon^S - \varepsilon^D}{[1 - (1-\alpha)\tau](\varepsilon^S - \varepsilon^D)} \right].\end{aligned}$$

Next, let us consider $p_D = (1 + \alpha\theta\tau)p$. Totally differentiating and holding τ constant, we obtain

$$\begin{aligned}dp_D &= (1 + \alpha\theta\tau)dp + \theta\tau p d\alpha + \alpha\tau p d\theta \\ \Leftrightarrow \frac{dp_D}{p_D} &= \frac{1 + \alpha\theta\tau}{p_D} dp + \theta\tau \frac{p}{p_D} d\alpha + \alpha\tau \frac{p}{p_D} d\theta \\ \Leftrightarrow \frac{dp_D}{d\alpha} \frac{1}{p_D} &= \frac{dp}{d\alpha} \frac{1}{p} + \frac{1}{1 + \alpha\theta\tau} \theta\tau + \alpha\tau \frac{1}{1 + \alpha\theta\tau} \frac{d\theta}{d\alpha} \\ &= -\tau \left[\frac{\frac{\varepsilon^S}{1-(1-\alpha)\tau} - \frac{\varepsilon^D}{1+\alpha\theta\tau} \left(\theta + \alpha \frac{d\theta}{d\alpha} \right)}{\varepsilon^S - \varepsilon^D} - \frac{1}{1 + \alpha\theta\tau} \left(\theta + \alpha \frac{d\theta}{d\alpha} \right) \right] \\ &= -\tau \left[\frac{\frac{1+\alpha\theta\tau}{1-(1-\alpha)\tau} \varepsilon^S - \varepsilon^S \left(\theta + \alpha \frac{d\theta}{d\alpha} \right)}{(1 + \alpha\theta\tau)(\varepsilon^S - \varepsilon^D)} \right].\end{aligned}$$

Under full salience ($\theta = 1$ and $\frac{d\theta}{d\alpha} = 0$), we obtain result (5). □

C Payroll taxes in OECD countries

Figure C.1: Payroll tax rates of employer and employee in OECD countries

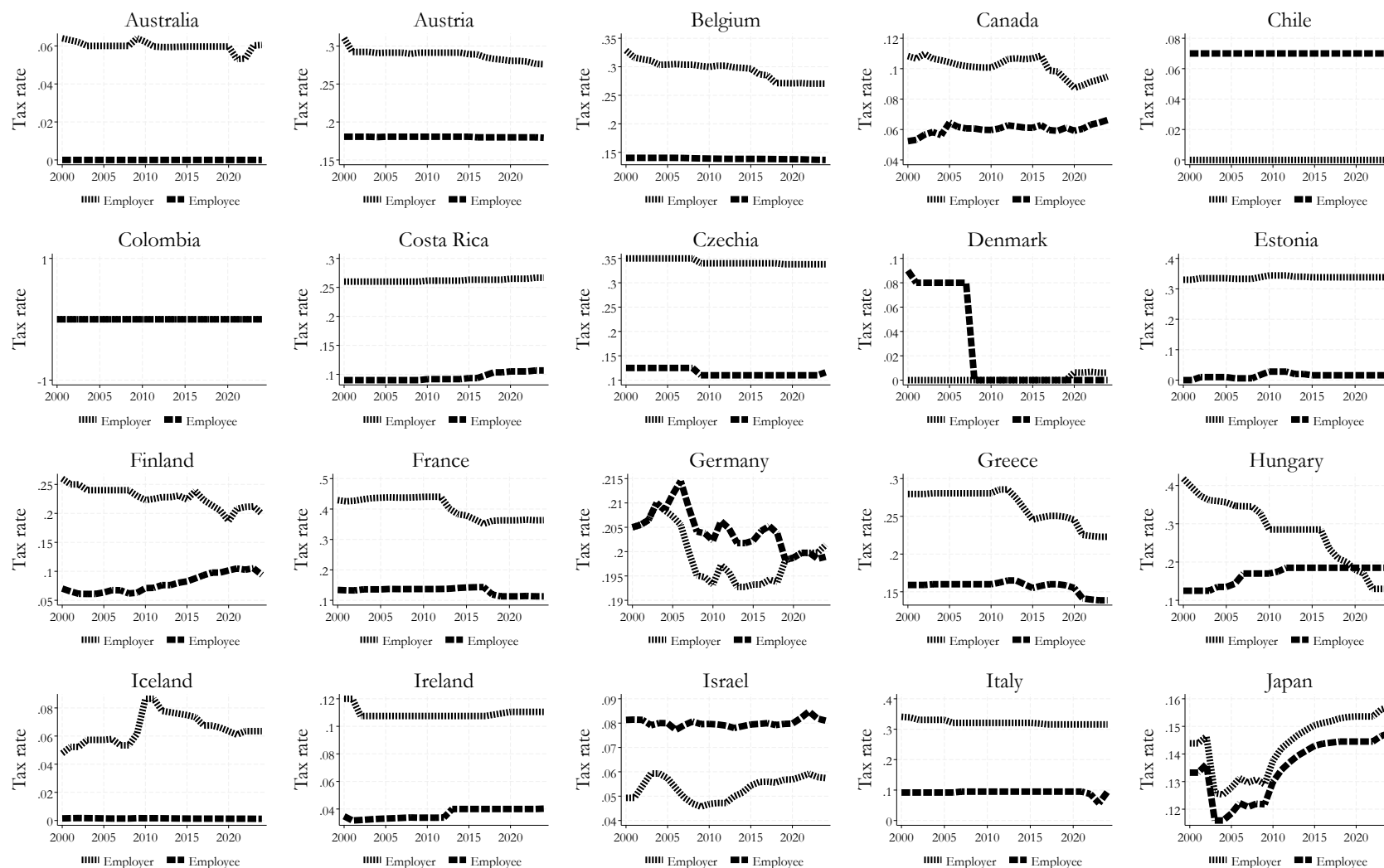
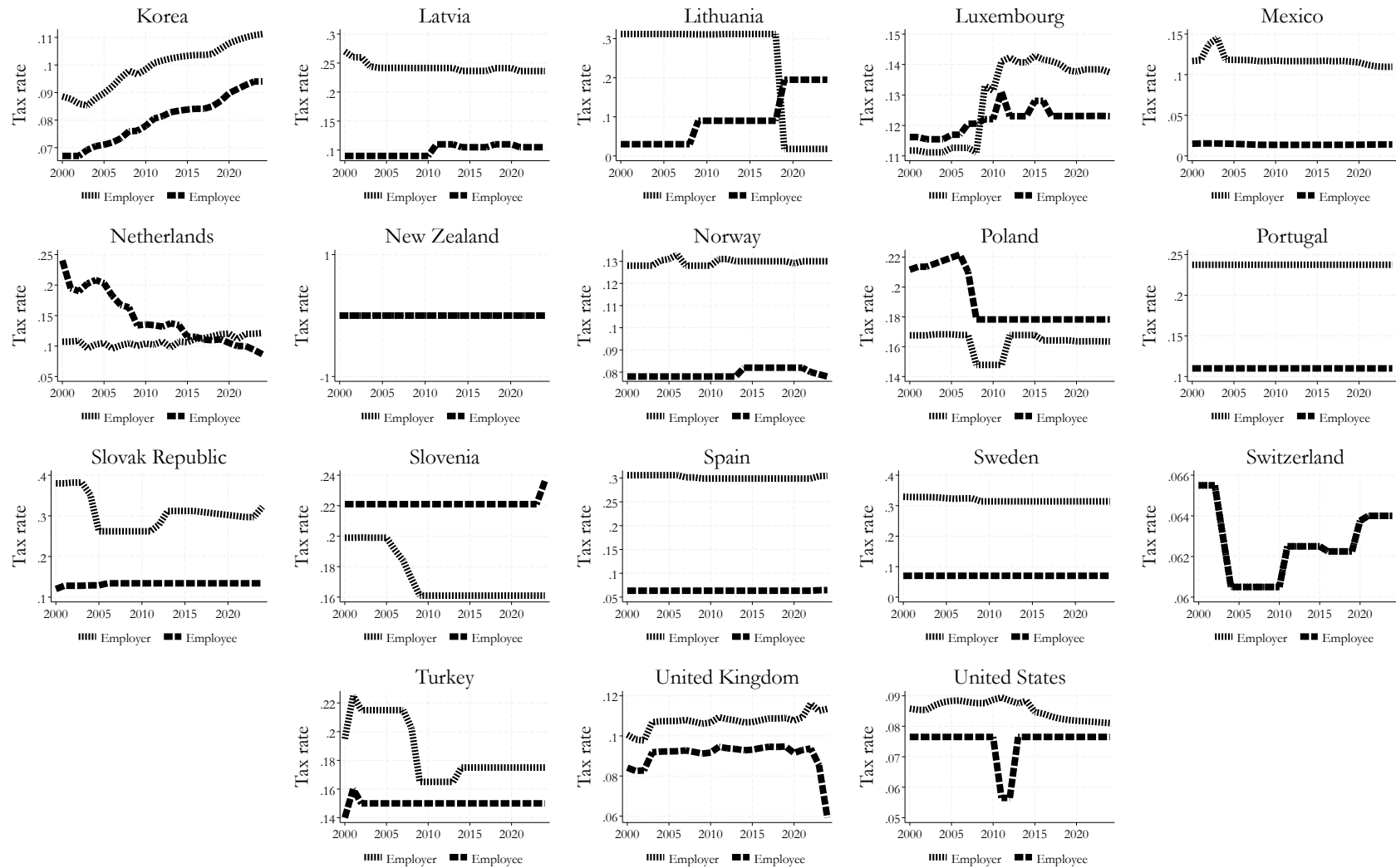


Figure C.1: Payroll tax rates of employer and employee in OECD countries



Notes: This figure shows the evolution of statutory employer and employee tax rates from 2000 to 2024 by all 38 OECD countries. The vertically dashed line corresponds to the employer side. The squared dashed line corresponds to the employee side. Based on *OECD Taxing Wages* data.